Gavin's Lectures on Physics - Cosmology



Author：长颈鹿 Gavin

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1 Introduction

This note draws primarily on "Modern Cosmology" (Second Edition) by Scott Dodelson.[2]

Einstein Field Equation:"Matter tells space-time how to curve."
Boltzmann Equation:"Space-time tells matter how to move."

Einstein Field Equation 0th Order: Friedmann Equation.
Boltazmann Equation 0th Order: Particle Evolution Equation.

Einstein Field Equation 1st Order: Poisson Equation $\leftrightarrow$ Newton's Gravitational Law.
Boltzmann Equation 1st Order: 1.Continuity Equation $\leftrightarrow$ Matter Conservation, 2.Euler Equation $\leftrightarrow$ Newton's 2nd Law

2 Expanding Universe

2.1 Structure in the Universe

Matter Power Spectrum:

$$\langle \delta(\mathbf{k_1})\delta(\mathbf{k_2})\rangle = (2\pi)^3 \delta_D^{(3)}(\mathbf{k_1}+\mathbf{k_2})P(k_1) \tag{1}$$

Correlation Function:

$$\xi(\mathbf{x}-\mathbf{x}') = \langle \delta(\mathbf{x_1})\delta(\mathbf{x_2})\rangle = \int \frac{d^3k}{(2\pi)^3} P(k) e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \tag{2}$$

2.2 Entropy Density

$$s = \frac{\rho + P}{T} \propto \frac{1}{a^3} \tag{3}$$

$$s = \frac{2\pi^2}{45}\Big(\sum_{i=bosons} g_i T_i^3 + \frac{7}{8}\sum_{i=fermions} g_i T_i^3\Big) \tag{4}$$

2.3 Cosmic Inventory

Photons: $\Omega_{\gamma 0}h^2 = 2.47 \times 10^{-5}$
Baryons: $\Omega_{b0}h^2 = 0.0225$
Matter: $\Omega_{m0}h^2 = 0.1431$
Neutrinos: $\Omega_{\nu 0}h^2 = \sum_i m_{\nu i}/94eV$

Distribution of Fermions and Bosons:

$$f_{FD}(E) = \frac{1}{e^{(E-\mu)/T}+1}, \quad f_{BE}(E) = \frac{1}{e^{(E-\mu)/T}-1} \tag{5}$$

Density:

$$\rho_s = \begin{cases} m_s n_s & ,\text{Matter} \\ g_s \frac{\pi^2}{30} T^4 & ,\text{Radiation Bosons} \\ \frac{7}{8} g_s \frac{\pi^2}{30} T^4 & ,\text{Radiation Fermions} \end{cases} \tag{6}$$

$$\Omega_{s0} = \frac{\rho_{s0}}{\rho_{cr0}} \tag{7}$$

Temperature:

$$T_s \propto \begin{cases} \frac{1}{a} & ,(m_s << T) \\ \frac{1}{a^2} & ,(m_s >> T) \end{cases} \tag{8}$$

2.4 Distances

Define a special function for different curvature K:

$$\sin_k(\chi) = \begin{cases} K^{-1/2}\sin(\sqrt{K}\chi), & K > 0 \\ \chi, & K = 0 \\ (-K)^{-1/2}\sinh(\sqrt{-K}\chi), & K < 0 \end{cases} \tag{9}$$

Comoving Distance:

$$\chi(t) = \int_t^{t_0} \frac{dt'}{a(t')} = \int_a^1 \frac{da_1}{a_1^2 H(a_1)} = \int_0^z \frac{dz_1}{H(z_1)} \tag{10}$$

Angular Diameter Distance:

$$d_A = \frac{l}{\theta} = a\sin_k(\chi) = \frac{\sin_k(\chi)}{1+z} \tag{11}$$

Luminosity Distance:

$$d_L = \frac{\sin_k(\chi)}{a} = (1+z)\sin_k(\chi) \tag{12}$$

Flux:

$$F = \frac{La^2}{4\pi\sin_k^2(\chi)(a)} = \frac{L}{4\pi d_L^2} \tag{13}$$

3 Einstein Field Equation

Einstein Field Equation:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G T_{\mu\nu} \tag{14}$$

Connection:

$$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu} \left(g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu} \right) \tag{15}$$

Ricci Tensor:

$$R_{\mu\nu} = \Gamma^{\alpha}_{\mu\nu,\alpha} - \Gamma^{\alpha}_{\mu\alpha,\nu} + \Gamma^{\alpha}_{\beta\alpha}\Gamma^{\beta}_{\mu\nu} - \Gamma^{\alpha}_{\beta\nu}\Gamma^{\beta}_{\mu\alpha} \tag{16}$$

Geodesic Equation:

$$\frac{d^2 x^{\mu}}{d\lambda^2} + \Gamma^{\mu}_{\alpha\beta} \frac{dx^{\alpha}}{d\lambda} \frac{dx^{\beta}}{d\lambda} = 0 \tag{17}$$

4 Boltzmann Equation

Boltzmann Equation:

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x^i}\frac{dx^i}{dt} + \frac{\partial f}{\partial p}\frac{dp}{dt} + \frac{\partial f}{\partial \hat{p}^i}\frac{d\hat{p}^i}{dt} \tag{18}$$

For reaction: $1 + 2 \longleftrightarrow 3 + 4$

Collision Term:

$$\begin{aligned} C[f_1(\mathbf{p})] = & \frac{1}{2E_1(p)} \int \frac{d^3q}{(2\pi)^3 2E_2(q)} \int \frac{d^3p'}{(2\pi)^3 2E_3\left(p'\right)} \int \frac{d^3q'}{(2\pi)^3 2E_4\left(q'\right)} \\ & \times |M|^2 (2\pi)^4 \delta_0^{(3)}\left(\mathbf{p} + \mathbf{q} - \mathbf{p}' - \mathbf{q}'\right) \delta_D^{(1)}\left(E_1 + E_2 - E_3 - E_4\right) \\ & \times \{f_3(\mathbf{p}') f_4(\mathbf{q}') \left[1 \pm f_1(\mathbf{p})\right] \left[1 \pm f_2(\mathbf{q})\right] - f_1(\mathbf{p}) f_2(\mathbf{q}) \left[1 \pm f_3(\mathbf{p}')\right] \left[1 \pm f_4(\mathbf{q}')\right]\} \end{aligned} \tag{19}$$

Here, + for bosons, - for fermions

5 Einstein Field Equation: 0th Order

5.1 Continuity Equation

From:

$$\nabla_\mu T^\mu_\nu = \frac{\partial T^\mu_\nu}{\partial x^\mu} + \Gamma^\mu_{\alpha\mu} T^\alpha_\nu - \Gamma^\alpha_{\mu\nu} T^\mu_\alpha = 0 \tag{20}$$

We get:

$$\frac{\partial \rho}{\partial t} + 3H(\rho + P) = 0 \tag{21}$$

Set:

$$\omega_s = \frac{P_s}{\rho_s} \tag{22}$$

Result in:

$$\rho_s \propto a^{-3(1+\omega_s)} \tag{23}$$

For different gradients:

$$\omega_m = 0, \omega_r = \frac{1}{3}, \omega_K = -\frac{1}{3}, \omega_\Lambda = -1 \tag{24}$$

5.2 FLRW metric

FLRW metric:

$$ds^2 = -dt^2 + a^2(t)[d\chi^2 + \sin_k^2(\chi)(d\theta^2 + \sin^2\theta d\phi^2)] \tag{25}$$

Energy-Momentum Tensor:

$$T^\mu_\nu = \begin{pmatrix} -\rho & & & \\ & P & & \\ & & P & \\ & & & P \end{pmatrix} \tag{26}$$

5.3 0th Order Einstein Field Equation (Friedmann Equation)

Friedmann equations:

$$\begin{aligned} (\frac{\dot{a}}{a})^2 + \frac{K}{a^2} &= \frac{8\pi G}{3}\rho \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3P) \end{aligned} \tag{27}$$

Friedmann Equations can also be expressed as:

$$\begin{aligned} (\frac{\dot{a}}{a})^2 = H^2(a) &= H_0^2(\frac{\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} + \frac{\Omega_{K0}}{a^2} + \Omega_{\Lambda 0}) \\ \frac{\ddot{a}}{a} &= -\frac{H_0^2}{2}(\frac{2\Omega_{r0}}{a^4} + \frac{\Omega_{m0}}{a^3} - 2\Omega_{\Lambda 0}) \end{aligned} \tag{28}$$

Critical density:

$$\rho_{cr0} = \frac{3H_0^2}{8\pi G} \tag{29}$$

Curvature density fraction:

$$K = -H_0^2 \Omega_{K0} \tag{30}$$

Curvature radius:

$$r_K = |K|^{-1/2} \tag{31}$$

Parameter values:

$$\begin{gathered} K = 0 \\ H_0 = 100h \cdot km \cdot s^{-1} \cdot Mpc^{-1} \\ a_{eq} = \frac{\Omega_{r0}}{\Omega_{m0}} = 2.9 \times 10^{-4} \end{gathered} \tag{32}$$

For dynamical dark energy:

$$\Omega_{\Lambda 0} \to \Omega_{DE} \exp[3 \int_a^1 (1 + \omega(a)) \frac{da}{a}] \tag{33}$$

6 Boltzmann Equation: 0th Order

6.1 0th Order Boltzmann Equation

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{p}{E}\frac{\hat{p}^i}{a}\frac{\partial f}{\partial x^i} - Hp\frac{\partial f}{\partial p} \tag{34}$$

$$\frac{dn(t)}{dt} + 3Hn(t) = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{35}$$

$$a^{-3}\frac{d(na^3)}{dt} = \int \frac{d^3p}{(2\pi)^3} C[f] \tag{36}$$

6.2 For Reaction at Low Temperature

Reaction $1 + 2 \longleftrightarrow 3 + 4$ when $T << E - \mu$:

$$a^{-3}\frac{d(n_1 a^3)}{dt} = n_1^{(0)} n_2^{(0)} \langle \sigma v \rangle \{ \frac{n_3 n_4}{n_3^{(0)} n_4^{(0)}} - \frac{n_1 n_2}{n_1^{(0)} n_2^{(0)}} \} \tag{37}$$

Number density:

$$n_s = n_s^{(0)} e^{\mu_s / T_s} = g_s \int \frac{d^3p}{(2\pi)^3} f(E) \tag{38}$$

Number density for $\mu_s = 0$:

$$n_s^{(0)} = \begin{cases} g_s (\frac{m_s T_s}{2\pi})^{\frac{3}{2}} e^{-m_s/T_s}, (m_s >> T) \\ g_s \frac{T_s^3}{\pi^2}, (m_s << T) \end{cases} \tag{39}$$

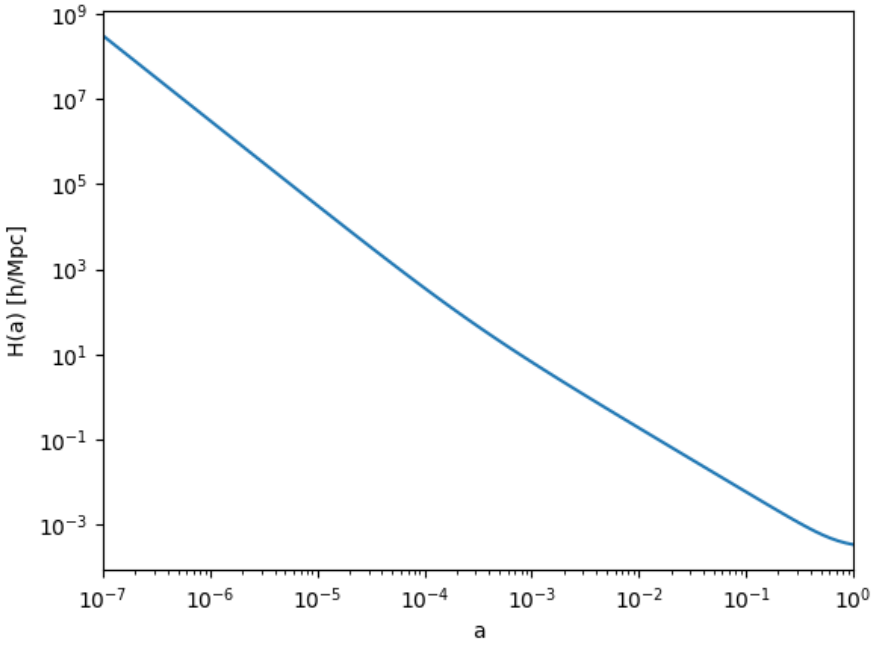


Figure 1: Hubble parameter H(a)

7 Homogeneous Results

7.1 Friedmann Equation Results

Age of the universe:

$$t = \int_0^1 \frac{da}{aH(a)} = 13.8\, billion\ years \tag{40}$$

Radius of visible universe:

$$r_{vi} = \int_0^1 \frac{da}{a^2 H(a)} = 46.3\, billion\ light\ years \tag{41}$$

Hubble parameter in Figure 1
Expansion rate $\dot{a} = H(z)/(1+z)$ in Figure 2 or Figure 3

7.2 Big Bang Nuclear Synthesis

Reaction between proton and neutron:

$$p + e^- \leftrightarrow n + \nu_e \tag{42}$$

Result in Figure 4

7.3 Recombination

Reaction at recombination:

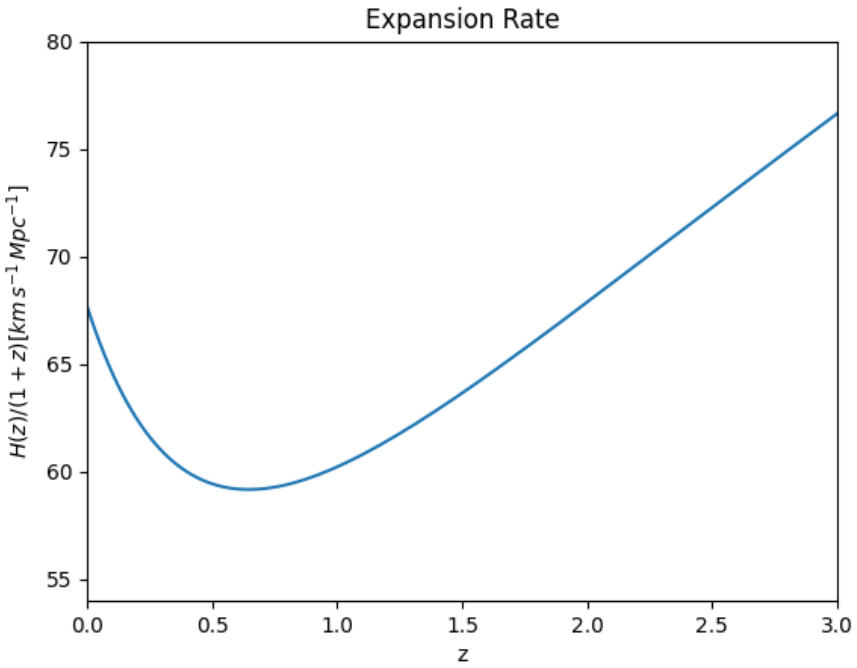


Figure 2: Expansion Rate

$$p + e^- \leftrightarrow H + \gamma \tag{43}$$

Fraction of free electrons:

$$X_e = \frac{n_e}{n_e + n_H} \tag{44}$$

X_e in Figure 5

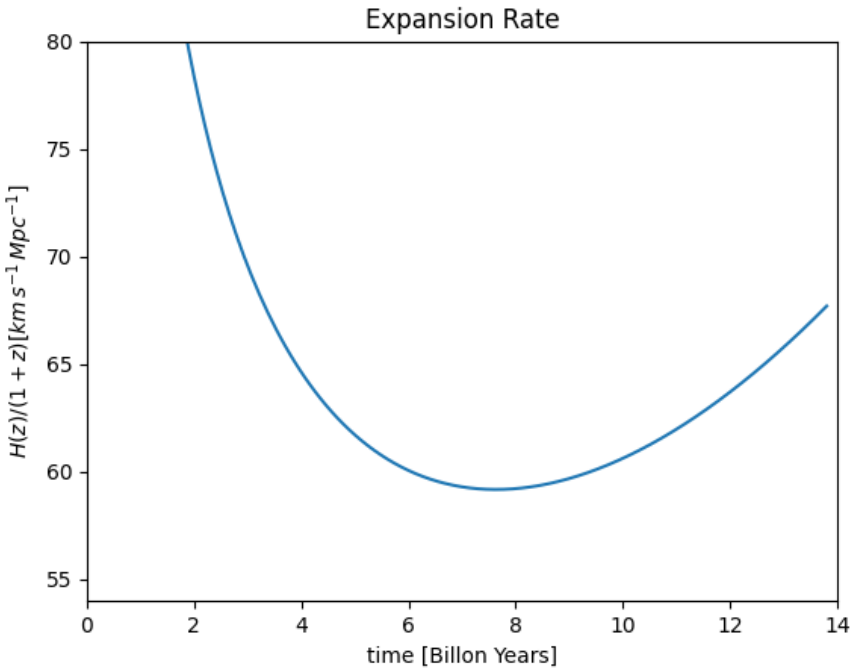


Figure 3: Expansion Rate

8 Einstein Field Equation: 1st Order

8.1 Metric

Scalar Perturbation:

$$\begin{cases} g_{00} = -1 - 2\Psi \\ g_{0i} = 0 \\ g_{ij} = a^2(t)\delta_{ij}(1 + 2\Phi) \end{cases} \tag{45}$$

Tensor Perturbation:

$$\begin{cases} \delta g_{ij} = a^2(t) h_{ij}^{TT} \\ h_{ij}^{TT} = \begin{pmatrix} h_+ & h_\times & 0 \\ h_\times & -h_+ & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{cases} \tag{46}$$

8.2 Energy-Momentum Tensor

$$\begin{aligned} T_\nu^\mu(\mathbf{x}, t) &= g \int \frac{d^3p}{(2\pi)^3} \frac{p^\mu p_\nu}{E(p)} f(\mathbf{x}, \mathbf{p}, t) \\ p^\mu &= [E(1 - \Psi), p^i(\frac{1 - \Phi}{a})] \\ -m^2 &= g_{\mu\nu} p^\mu p^\nu \end{aligned} \tag{47}$$

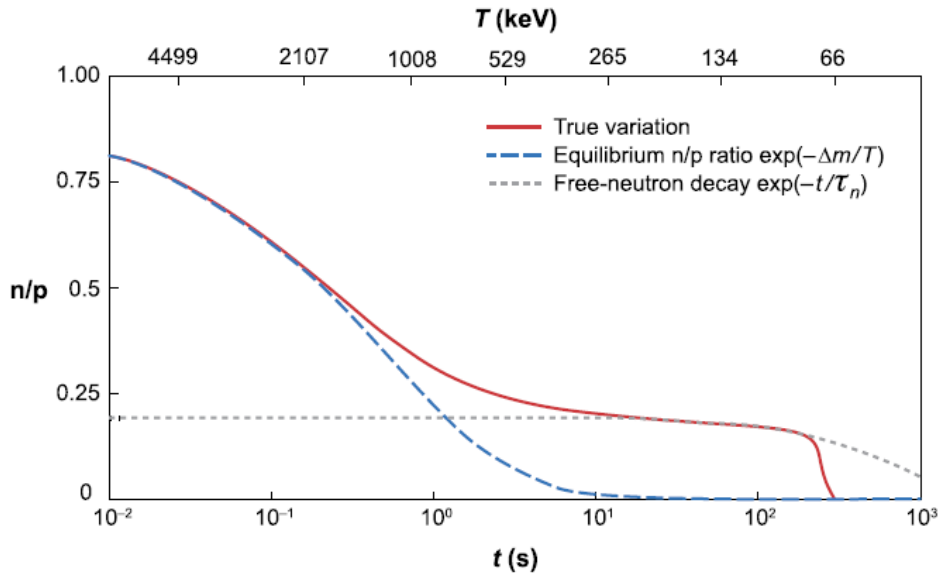


Figure 4: BBN

$$
\begin{aligned}
T^0{}_0(x,t) &= -g\int \frac{d^3p}{(2\pi)^3}E(p)f(\mathbf{x},\mathbf{p},t)\\
T^0{}_i(x,t) &= ga(1+\Phi-\Psi)\int \frac{d^3p}{(2\pi)^3}p_i f(\mathbf{x},\mathbf{p},t)\\
T^i{}_j(x,t) &= g\int \frac{d^3p}{(2\pi)^3}\frac{p^i p_j}{E(p)}f(\mathbf{x},\mathbf{p},t)
\end{aligned}
\tag{48}
$$

8.3 1st Order Einstein Field Equation

Scalar Perturbation(Poisson's Equation):

$$
\begin{aligned}
k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0}]\\
k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\rm r}\Theta_{\rm r,2}\\
k^2\Phi = 4\pi\, G a^2&\left[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0} + \frac{3aH}{k}(i\rho_{\rm m}u_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,1})\right].
\end{aligned}
\tag{49}
$$

Tensor Perturbation:

$$
h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times) \tag{50}
$$

Matter(m) contains CDM(c) and Baryons(b).
Radiation(r) contains Photons(γ) and Neutrinos(ν).

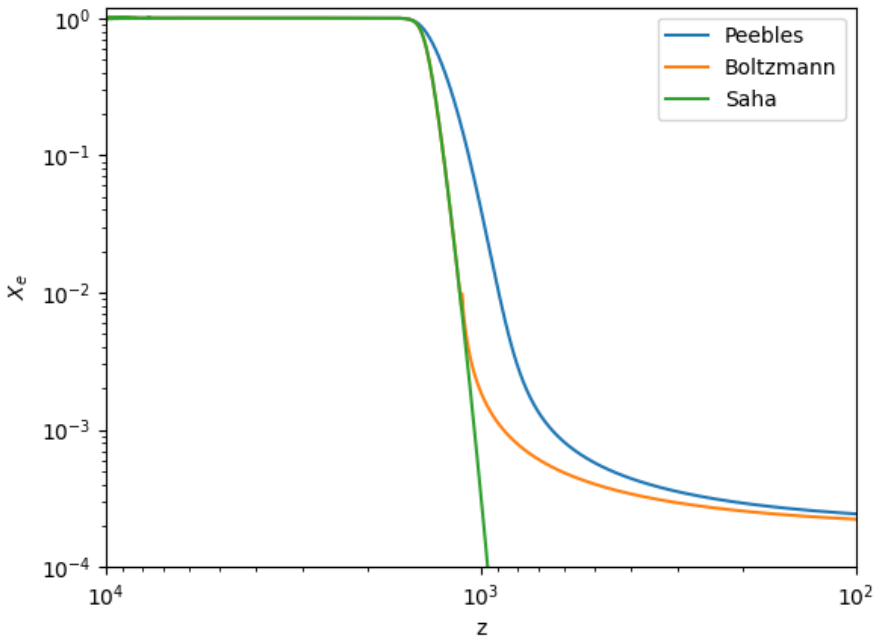


Figure 5: X_e

9 Boltzmann Equation: 1st Order

$$C[f] = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x^i}\frac{\hat{p}^i}{a}\frac{p}{E}(1 - \Phi + \Psi) - \frac{\partial f}{\partial p}\left\{\left[H + \dot{\Phi}\right]p + \frac{E}{a}\hat{p}^i\Psi_{,i}\right\} + \frac{\partial f}{\partial \hat{p}^i}\frac{E}{ap}\left[(\delta^{ik} - \hat{p}^i\hat{p}^k)(\frac{p^2}{E^2}\Phi - \Psi)_{,k}\right]. \tag{51}$$

9.1 Photons

9.1.1 Anisotropy

Temperature Perturbation:

$$\Theta = \delta T/T \tag{52}$$

Multipole:

$$\Theta_l = \frac{1}{(-i)^l}\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\Theta(\mu, k, \eta) \tag{53}$$

Optical depth:

$$\tau(\eta) = \int_{\eta}^{\eta_0} d\eta_1 n_e \sigma_T a \tag{54}$$

Visibility function:

$$g(\eta) = -\tau'(\eta)e^{-\tau(\eta)} \tag{55}$$

From visibility function, we get recombination at redshift about $z_{CMB} = 1080$ in Figure 6

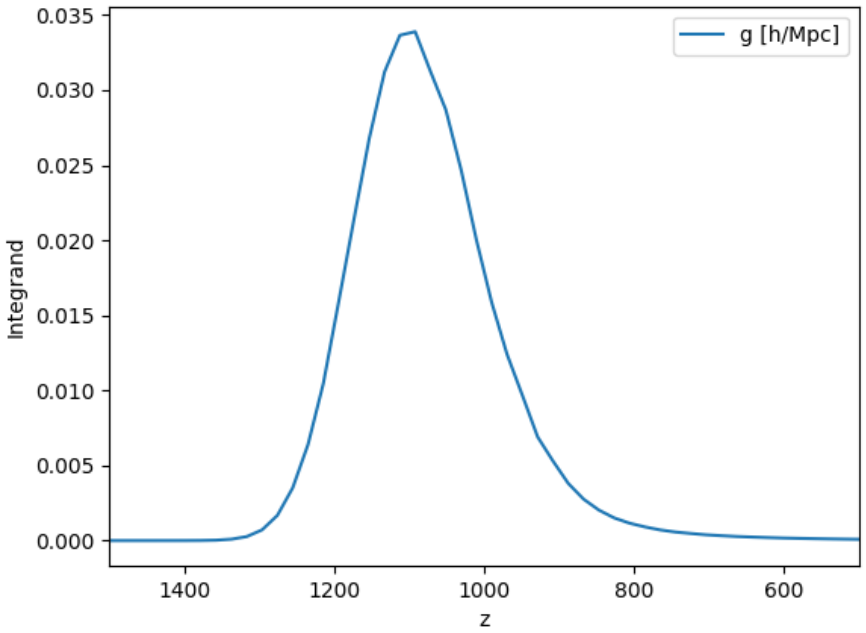


Figure 6: g(z)

Linear equation for photon anisotropy (CMB anisotropy):

$$\begin{aligned} \Theta' + ik\mu\Theta = -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi = \Theta_2 + \Theta_{P2} + \Theta_{P0} \end{aligned} \tag{56}$$

For each multipole:

$$\begin{aligned} l = 0: &\quad \Theta_0' + k\Theta_1 = -\Phi', \\ l = 1: &\quad \Theta_1' - \frac{1}{3}k(\Theta_0 - 2\Theta_2) = \frac{k\Psi}{3} + \tau'\left[\Theta_1 - \frac{iu_{\rm b}}{3}\right]. \\ l = 2: &\quad \Theta_2' - \frac{1}{5}k(2\Theta_1 - 3\Theta_3) = (\frac{9}{10}\Theta_2 - \frac{\Theta_{P0}}{10} - \frac{\Theta_{P2}}{10})\tau' \\ l > 2: &\quad \Theta_l' + \frac{k}{(-i)^{l+1}}\int_{-1}^{1}\frac{d\mu}{2}\mu\mathcal{P}_l(\mu)\Theta(\mu) = \tau'\Theta_l \end{aligned} \tag{57}$$

9.1.2 Polarization

Linear equation for photon polarization (CMB polarization, E mode):

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E = \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi = \Theta_2 + \Theta_{E2} + \Theta_{E0} \end{aligned} \tag{58}$$

9.2 Cold Dark Matter(CDM)

CDM perturbation:

$$\delta_c = \delta n_c/\bar{n} \tag{59}$$

Linear Equation for CDM:

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad \text{(Euler's Equation)} \end{aligned} \tag{60}$$

9.3 Baryons

Baryon perturbation:

$$\delta_b = \delta\rho_e/\rho_e = \delta\rho_p/\rho_p \tag{61}$$

Linear Equation for Baryons:

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad \text{(Continuity Equation)} \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad \text{(Euler's Equation)} \end{aligned} \tag{62}$$

The ratio of baryon to photon density:

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{63}$$

9.4 Neutrinos

Neutrino perturbation:

$$\mathcal{N} = \delta T_\nu/T_\nu \tag{64}$$

Linear Equation for Neutrinos:

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{65}$$

10 Initial Condition

10.1 Slow Roll Inflation Model

Scalar Inflation Field:

$$\begin{aligned} \rho &= \frac{1}{2}\dot{\phi}^2 + V(\phi) \\ \mathcal{P} &= \frac{1}{2}\dot{\phi}^2 - V(\phi) \end{aligned} \tag{66}$$

Slow Roll Variables:

$$\begin{aligned} \epsilon_{\rm sr} &\equiv \frac{d}{dt}\left(\frac{1}{H}\right) \\ \delta_{\rm sr} &\equiv \frac{1}{H}\frac{\ddot{\phi}}{\dot{\phi}} \end{aligned} \tag{67}$$

10.2 Initial Perturbations

$$\begin{aligned} \Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\ \Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\ \delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\ \Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\ iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R} \end{aligned} \tag{68}$$

10.3 Initial Power Spectrum

Initial Power Spectrum:

$$\begin{aligned} P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2\epsilon_{\rm Sr}}\bigg|_{aH=k} \equiv 2\pi^2\mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm P}}\right)^{n_s-1} \\ P_{\rm T}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\rm T} k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_{\rm T}} \end{aligned} \tag{69}$$

Parameters:

$$\begin{aligned} n_s &= 1 - 4\epsilon_{\rm sr} - 2\delta_{\rm sr} = 0.9682 \\ n_{\rm T} &= -2\epsilon_{\rm sr} \approx 0 \\ k_p &= 0.05 Mpc^{-1} \\ ln(10^{10}A_s) &= 3.0480 \end{aligned} \tag{70}$$

Ratio of Initial Tensor and Scalar Power Spectrum:

$$r(k) \equiv \frac{P_{\rm T}(k)}{P_{\mathcal{R}}(k)} \overset{k=k_{\rm p}}{=} \frac{\mathcal{A}_{\rm T}}{\mathcal{A}_s} = 16\epsilon_{\rm sr}\Big|_{aH=k} \tag{71}$$

11 Linear Theory Summary

11.1 Linear Equations

Linear Equation for Scalar Perturbation:

$$\begin{aligned} k^2\Phi + 3\frac{a'}{a}\left(\Phi' - \Psi\frac{a'}{a}\right) &= 4\pi G a^2[\rho_{\rm m}\delta_{\rm m} + 4\rho_{\rm r}\Theta_{\rm r,0}](\text{Poisson's Equation}) \\ k^2(\Phi+\Psi) &= -32\pi G a^2\rho_{\rm r}\Theta_{\rm r,2} \end{aligned} \tag{72}$$

Linear Equation for Tensor Perturbation:

$$h'' + 2\frac{a'}{a}h_t' + k^2 h_t = 0 \quad ,(t=+,\times) \tag{73}$$

Linear Equation for Photon Anisotropy (CMB anisotropy):

$$\begin{aligned} \Theta' + ik\mu\Theta &= -\Phi' - ik\mu\Psi - \tau'\left[\Theta_0 - \Theta + \mu u_{\rm b} - \frac{1}{2}\mathcal{P}_2(\mu)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{P2} + \Theta_{P0}. \end{aligned} \tag{74}$$

Linear Equation for Photon Polarization (CMB polarization, E mode):

$$\begin{aligned} \Theta_E' + ik\mu\Theta_E &= \tau'\left[\Theta_E - \frac{3}{4}(1-\mu^2)\Pi\right] \\ \Pi &= \Theta_2 + \Theta_{E2} + \Theta_{E0}. \end{aligned} \tag{75}$$

Linear Equation for CDM:

$$\begin{aligned} \delta_c' + iku_c &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_c' + \frac{a'}{a}u_c &= -ik\Psi \quad (\text{Euler's Equation}) \end{aligned} \tag{76}$$

Linear Equation for Baryons:

$$\begin{aligned} \delta_b' + iku_b &= -3\Phi' \quad (\text{Continuity Equation}) \\ u_b' + \frac{a'}{a}u_b &= -ik\Psi + \frac{\tau'}{R_b}(u_b + 3i\Theta_1) \quad (\text{Euler's Equation}) \end{aligned} \tag{77}$$

Linear Equation for Neutrinos:

$$\mathcal{N}' + ik\mu\,\frac{p}{E_\nu(p)}\mathcal{N} - p\frac{a'}{a}\frac{\partial}{\partial p}\mathcal{N} = -\Phi' - ik\mu\frac{E_\nu(p)}{p}\Psi. \tag{78}$$

11.2 Initial Condition

Initial Condition:

$$
\begin{aligned}
\Psi = -\frac{2}{3}\mathcal{R}, \Phi = \frac{2}{3}\mathcal{R}, \\
\Theta_0 = \frac{1}{3}\mathcal{R}, \mathcal{N}_0 = \frac{1}{3}\mathcal{R} \\
\delta_b = \mathcal{R}, \delta_c = \mathcal{R} \\
\Theta_1 = -\frac{k}{9aH}\mathcal{R}, \mathcal{N}_1 = -\frac{k}{9aH}\mathcal{R} \\
iu_b = -\frac{k}{3aH}\mathcal{R}, iu_c = -\frac{k}{3aH}\mathcal{R}
\end{aligned}
\tag{79}
$$

Initial Power Spectrum:

$$
\begin{aligned}
P_{\mathcal{R}}(k) &= \frac{2\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2 \epsilon_{\rm Sr}}\bigg|_{aH=k} \equiv 2\pi^2 \mathcal{A}_s k^{-3}\left(\frac{k}{k_{\rm P}}\right)^{n_s-1} \\
P_{\rm T}(k) &= 4P_h(k) = \frac{32\pi}{k^3}\frac{H^2}{m_{\rm Pl}^2}\bigg|_{aH=k} \equiv 2\pi^2 A_{\rm T} k^{-3}\left(\frac{k}{k_{\rm p}}\right)^{n_{\rm T}}
\end{aligned}
\tag{80}
$$

12 Linear Results: Matter and Potential

Linear Matter Power Spectrum in Figure 7, Figure 8

$$P_L(k,a) = \left[\frac{2(k^2+3a^2H^2)}{5H_0^2\Omega_{m0}}T(k)D_+(a)\right]^2 P_{\mathcal{R}}(k) \tag{81}$$

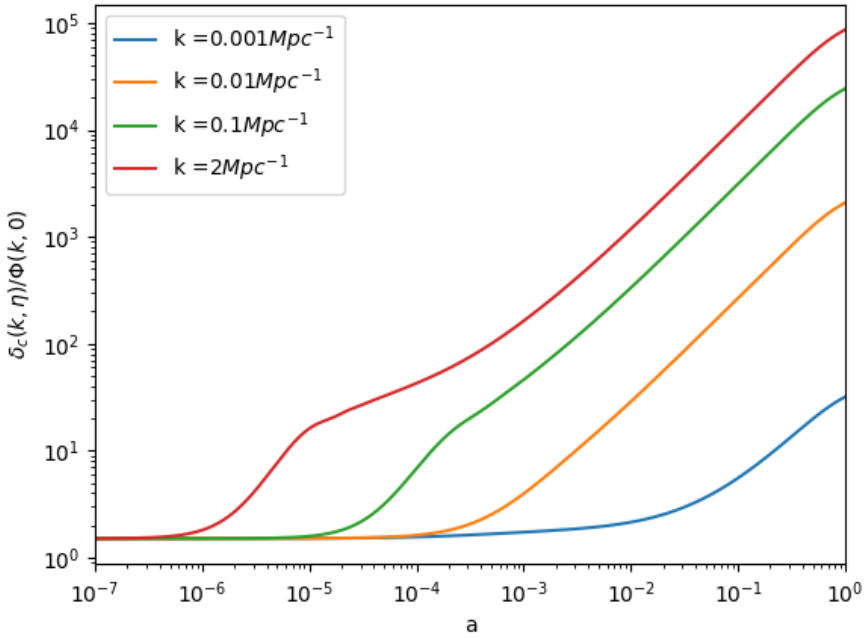


Figure 7: δ_c/Φ_0

Linear Potential Power Spectrum in Figure 9

$$P_\Phi(k,a) = \left[\frac{3}{5}T(k)\frac{D_+(a)}{a}\right]^2 P_{\mathcal{R}}(k) \tag{82}$$

Tensor perturbation in Figure 10. Smaller scale modes decay earlier.
Transfer function (BBKS):

$$\begin{aligned} T(k) = &\frac{ln(1+2.34\cdot q)}{2.34\cdot q} \\ &\cdot\left[1+(3.89\cdot q)+(16.1\cdot q)^2+(5.46\cdot q)^3+(6.71\cdot q)^4\right]^{-1/4} \end{aligned} \tag{83}$$

$$q(k) = \frac{k}{\Omega_{m0}h\cdot h/Mpc} \tag{84}$$

Growth Rate:

$$f(a) = \frac{dlnD_+(a)}{dlna} = \left[\frac{H_0^2}{H^2(a)}\frac{\Omega_{m0}}{a^3}\right]^{0.55} \tag{85}$$

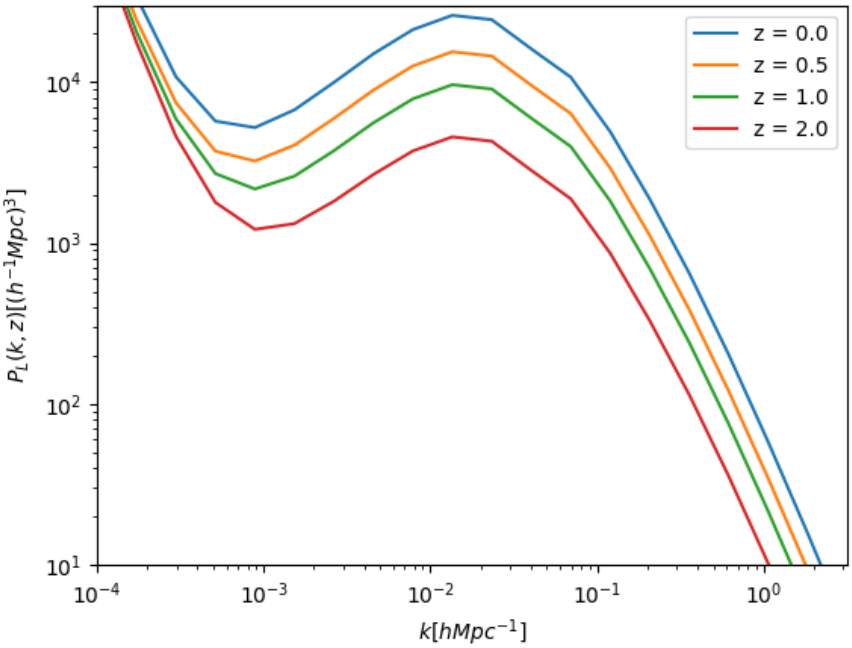


Figure 8: P_L

13 Linear Results: Photons (CMB)

13.1 Anisotropy

Cosmic Microwave Background(CMB) anisotropic power spectrum in Figure 11

$$C(l) = \frac{2}{\pi}\int_0^\infty dk k^2 P_{\mathcal{R}}(k)|T_l(k)|^2 \tag{86}$$

$$T_l(k) = \Theta_l(k,\eta_0)/\mathcal{R}(k) \tag{87}$$

The ratio of baryon to photon density:

$$R_b(\eta) = \frac{3\rho_b(\eta)}{4\rho_\gamma(\eta)} \tag{88}$$

Sound speed of baryon-photon fluid:

$$c_s(\eta) = \sqrt{\frac{1}{3[1+R_b(\eta)]}} \tag{89}$$

Sound horizon:

$$r_s(\eta) = \int_0^\eta d\tilde{\eta} c_s(\tilde{\eta}) \tag{90}$$

Multi-poles of Photons:

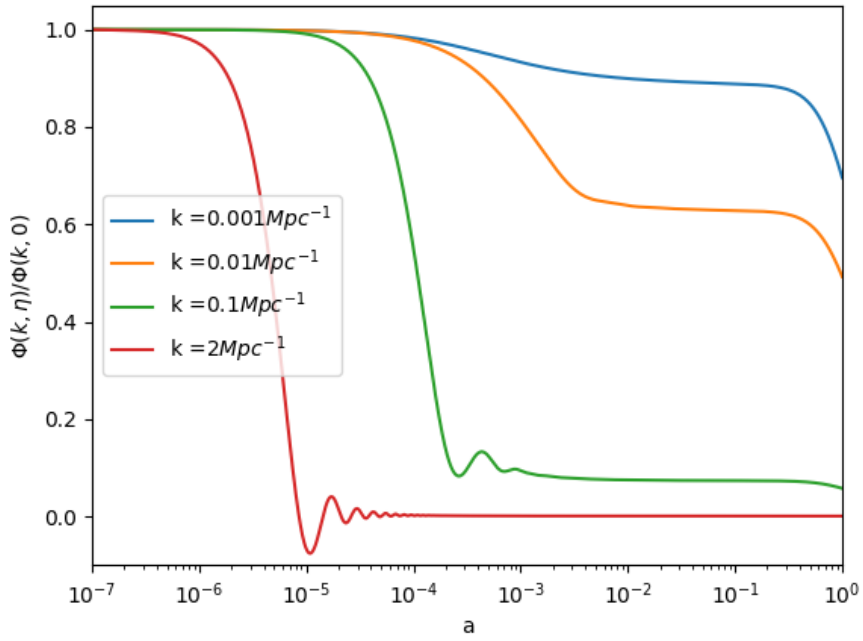


Figure 9: Φ/Φ_0

$$\begin{aligned}\Theta_l(k,\eta_0) \simeq & \left[\Theta_0(k,\eta_*)+\Psi(k,\eta_*)\right] j_l\left[k(\eta_0-\eta_*)\right] \\ & + 3\Theta_1(k,\eta_*)\left(j_{l-1}\left[k(\eta_0-\eta_*)\right]-(l+1)\frac{j_l\left[k(\eta_0-\eta_*)\right]}{k(\eta_0-\eta_*)}\right) \\ & + \int_0^{\eta_0} d\eta e^{-\tau}\left[\Psi'(k,\eta)-\Phi'(k,\eta)\right] j_l\left[k(\eta_0-\eta)\right]\end{aligned} \tag{91}$$

Monopole and Dipole:

$$\begin{aligned}\Theta_0(\mathbf{k},\eta) = & \{\left[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)\right]\cos(kr_s)-\Phi(\mathbf{k},\eta) \\ & + \frac{k}{\sqrt{3}}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\sin[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{92}$$

$$\begin{aligned}\Theta_1(\mathbf{k},\eta) = & \{\frac{1}{\sqrt{3}}[\Theta_0(\mathbf{k},0)+\Phi(\mathbf{k},0)]\sin(kr_s) \\ & - \frac{k}{3}\int_0^{\eta} d\tilde{\eta}[\Phi(\mathbf{k},\tilde{\eta})-\Psi(\mathbf{k},\tilde{\eta})]\cos[k(r_s(\eta)-r_s(\tilde{\eta}))]\}\cdot\exp(-\frac{k^2}{k_D^2})\end{aligned} \tag{93}$$

Monopole and Dipole in Wayne Hu's approximation (arxiv:9504057):

$$\frac{\Theta_0(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)\cos(kr_s(\eta))+(\mathcal{R}_b+1)\frac{\Phi(k,\eta)}{\mathcal{R}(k)}\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{94}$$

$$\frac{\Theta_1(k,\eta)}{\mathcal{R}(k)} = \left[-\frac{1}{5}(1+3\mathcal{R}_b)c_s\cdot\sin(kr_s(\eta))\right]\cdot\exp(-\frac{k^2}{k_D^2}) \tag{95}$$

Diffusion Damping:

$$\frac{1}{k_D^2(\eta)} \equiv \int_0^{\eta}\frac{d\tilde{\eta}}{6(1+R_b)n_e\sigma_{\mathrm{T}}a(\tilde{\eta})}\left[\frac{R_b^2}{1+R_b}+\frac{8}{9}\right] \tag{96}$$

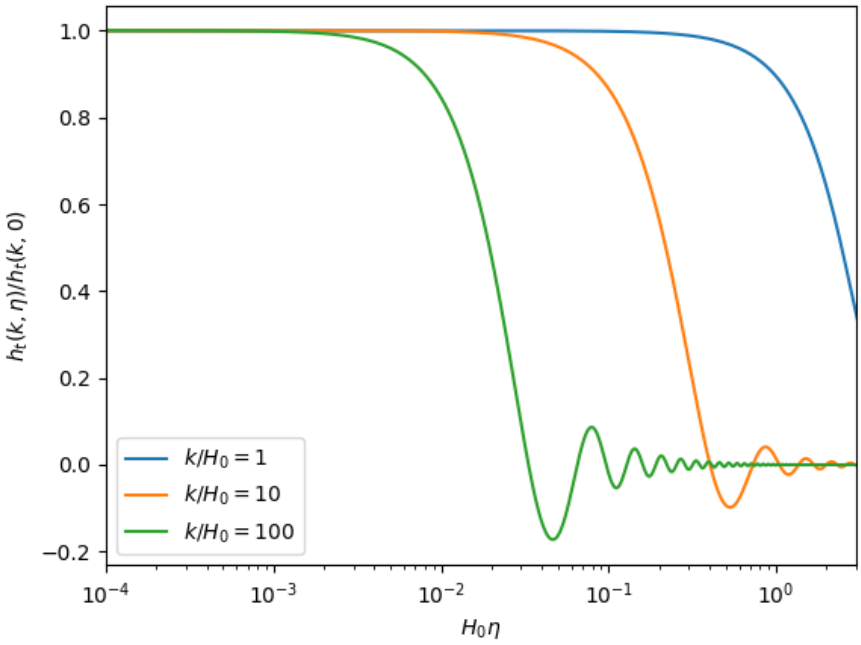


Figure 10: Tensor Perturbation

13.2 Polarization

(1) E and B modes
Polarization Tensor:

$$I_{ij} = \begin{pmatrix} I+Q & U \\ U & I-Q \end{pmatrix} \tag{97}$$

Meaning of polarization components Q,U in Figure 12
Change Q,U to rotation-invariant polarization components E,B in Figure 13

$$\begin{aligned} \nabla^2 E &= \partial_i \partial_j (I_{ij} - I\delta_{ij}) \\ \nabla^2 B &= \epsilon_{ik} \partial_j \partial_k (I_{ij} - I\delta_{ij}) \end{aligned} \tag{98}$$

Here,

$$\epsilon_{ij} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{99}$$

In Fourier space:

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij}) I_{ij} \\ B(\mathbf{l}) &= \epsilon_{ik} \frac{l^j l^k}{l^2} I_{ij} \end{aligned} \tag{100}$$

Set l as:

$$\mathbf{l} = l(\cos\phi_l, \sin\phi_l) \tag{101}$$

We get relation of Q,U and E,B:

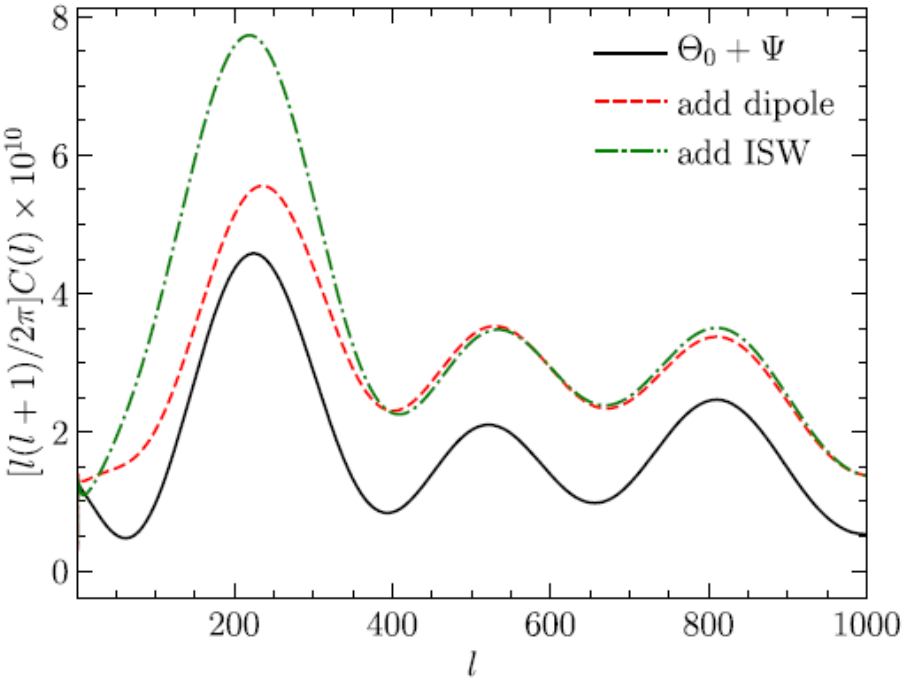


Figure 11: CMB Anisotropy

$$\begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ -\sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} \tag{102}$$

Reverse the matrix:

$$\begin{pmatrix} Q(\mathbf{l}) \\ U(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{103}$$

We can write polarization tensor I_{ij} in term of I, E and B:

$$I_{ij}(\mathbf{l}) = I(\mathbf{l})\delta_{ij} + \begin{pmatrix} \cos(2\phi_l) & \sin(2\phi_l) \\ \sin(2\phi_l) & -\cos(2\phi_l) \end{pmatrix} E(\mathbf{l}) + \begin{pmatrix} -\sin(2\phi_l) & \cos(2\phi_l) \\ \cos(2\phi_l) & \sin(2\phi_l) \end{pmatrix} B(\mathbf{l}) \tag{104}$$

(2) Power Spectrum
For different modes:

$$s = \{T, E, B\} \tag{105}$$

We have power spectrum in Figure 14

$$C_{s_1 s_2}(l) = \frac{2}{\pi} \int_0^\infty dk k^2 P_{\mathcal{R}}(k) |T_l^{s_1}(k) T_l^{s_2 *}(k)| \tag{106}$$

$$T_l^s(k) = \Theta_{s,l}(k, \eta_0)/\mathcal{R}(k) \tag{107}$$

E mode:

$$\Theta_{E,l}(k, \eta_0) = -\frac{5k}{6\tau'} \Theta_1(k, \eta_*) \frac{l^2}{(k\eta_0)^2} j_l(k\eta_0) \tag{108}$$

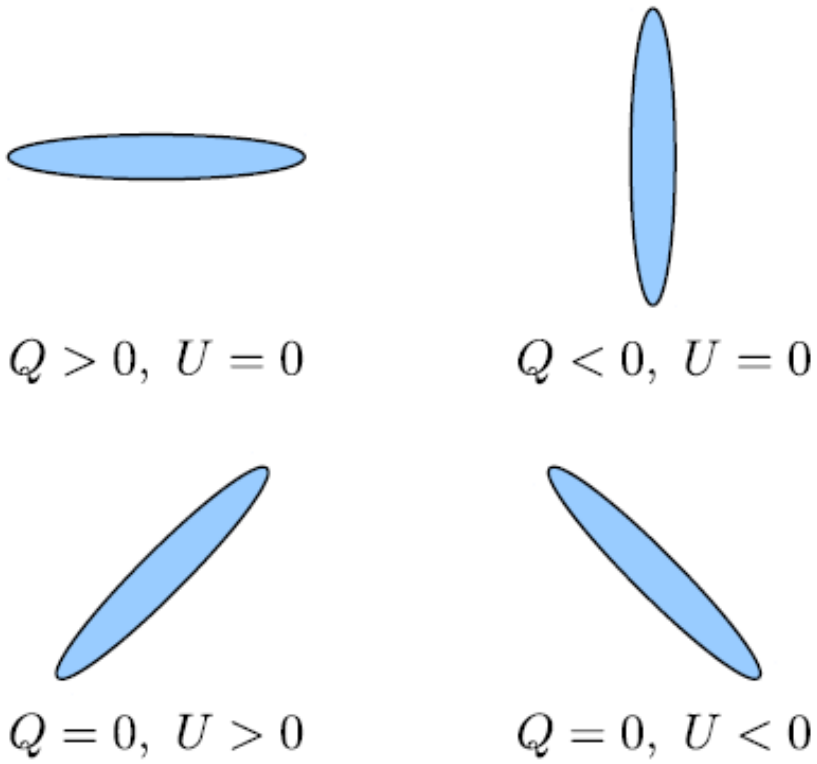


Figure 12: Polarization components:Q and U

B mode all comes from tensor perturbation:

$$B_{\times}^{T}(\mathbf{k}) \propto \frac{8\pi}{15} h_{\times}(\mathbf{k})(\hat{n} \cdot \hat{k}) \tag{109}$$

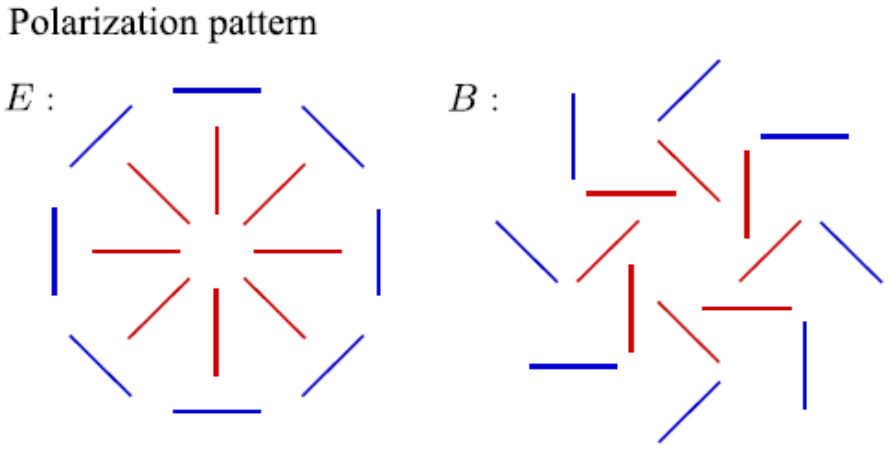


Figure 13: Rotation-invariant polarization components:E and B

14 Tracers

14.1 RSD + AP Effect

(1) Redshift space distortion (RSD) effect
RSD: Influence to power spectrum from peculiar velocity.

Galaxy bias:

$$\delta_g(\mathbf{x}, z) = b_1(z)\delta_m(\mathbf{x}, z) \tag{110}$$

Cosine of angle between the light of sight and the wave vector:

$$\mu_k = \widehat{\mathbf{e}}_z \cdot \widehat{\mathbf{k}} \tag{111}$$

(2) *Alcock − Paczyński* (AP) effect
AP effect: Influence to power spectrum from wrong cosmology.

Error of comoving distance and Hubble parameter between fiducial cosmology and real value:

$$\delta\chi = \chi_{fid} - \chi,\ \delta H = H_{fid} - H \tag{112}$$

Relation between true and observed $\mathbf{k}$:

$$\mathbf{k}[\mathbf{k}_{obs}] = ([1+\alpha_\perp]k^1_{\rm c,fid}, [1+\alpha_\perp]k^2_{\rm c,fid}, [1+\alpha_\parallel]k^3_{\rm c,fid}) \tag{113}$$

$$\alpha_\perp = \left.\frac{\delta\chi}{\chi_{fid}}\right|_{\bar{z}},\ \alpha_\parallel = -\left.\frac{\delta H}{H_{fid}}\right|_{\bar{z}} \tag{114}$$

(3) Observed galaxy power spectrum (RSD + AP effect):

$$P_{\rm g,obs}(\mathbf{k}_{\rm obs}, \bar{z}) = P_{\rm L}(k, \bar{z})\left[b_1 + f\mu_k^2\right]^2 + P_N \tag{115}$$

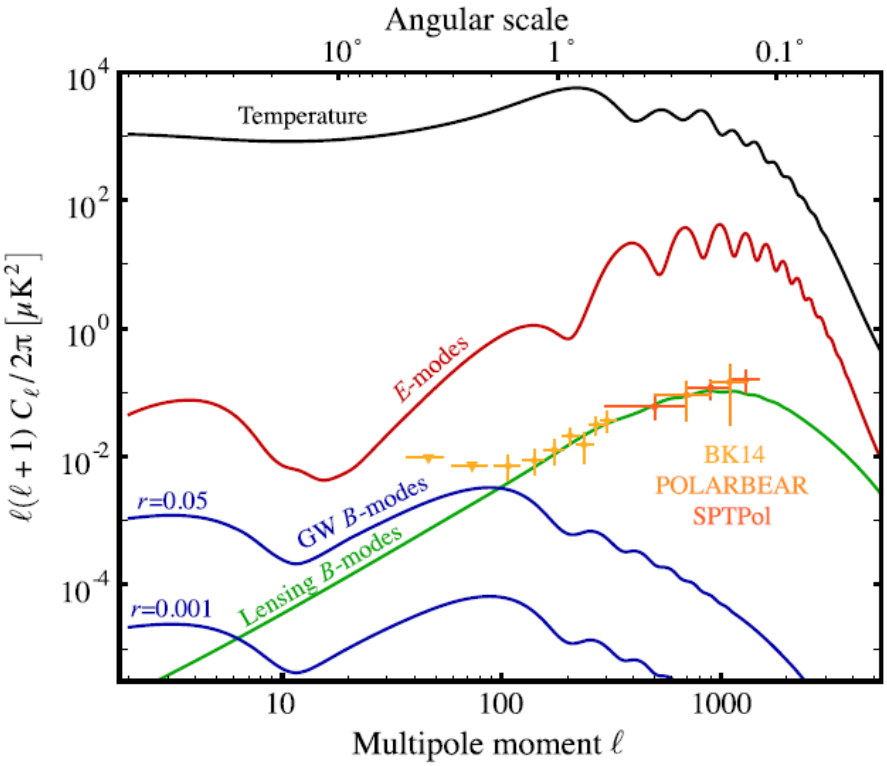


Figure 14: CMB Polarization

More accurate result, as shown in Figure 15:

$$P_{\mathrm{g,obs}}(\mathbf{k}_{\mathrm{obs}},\bar{z}) = P_{\mathrm{L}}(k,\bar{z})\left[b_1 + f\mu_k^2\right]^2 D^{FOG}(k\mu_k) + P_N \tag{116}$$

$$D^{FOG}(k_z = k\mu_k) = \left|\langle \exp(i\frac{k_z v_z}{H})\rangle\right|^2 \tag{117}$$

14.2 Angular Correlation

Distribution of galaxy distances:

$$W(\chi) = \frac{1}{N_g}\frac{dN_g}{d\chi} \tag{118}$$

Projected overdensity:

$$\Delta(\hat{\mathbf{n}}) = \int_0^\infty d\chi W(\chi)\delta_{g,obs}(\mathbf{x} = \hat{\mathbf{n}}\chi, \eta = \eta_0 - \chi) \tag{119}$$

Angular power spectrum:

$$\langle \Delta_{g,lm}\Delta^*_{g,l'm'}\rangle = \delta_{ll'}\delta_{mm'}C_g(l) \tag{120}$$

Angular power spectrum result (Limber approximation):

$$C_g(l) = \int \frac{d\chi}{\chi^2}W^2(\chi)P_{g,obs}\left(k = \frac{l+1/2}{\chi}, \mu_k = 0, \eta(\chi)\right) \tag{121}$$

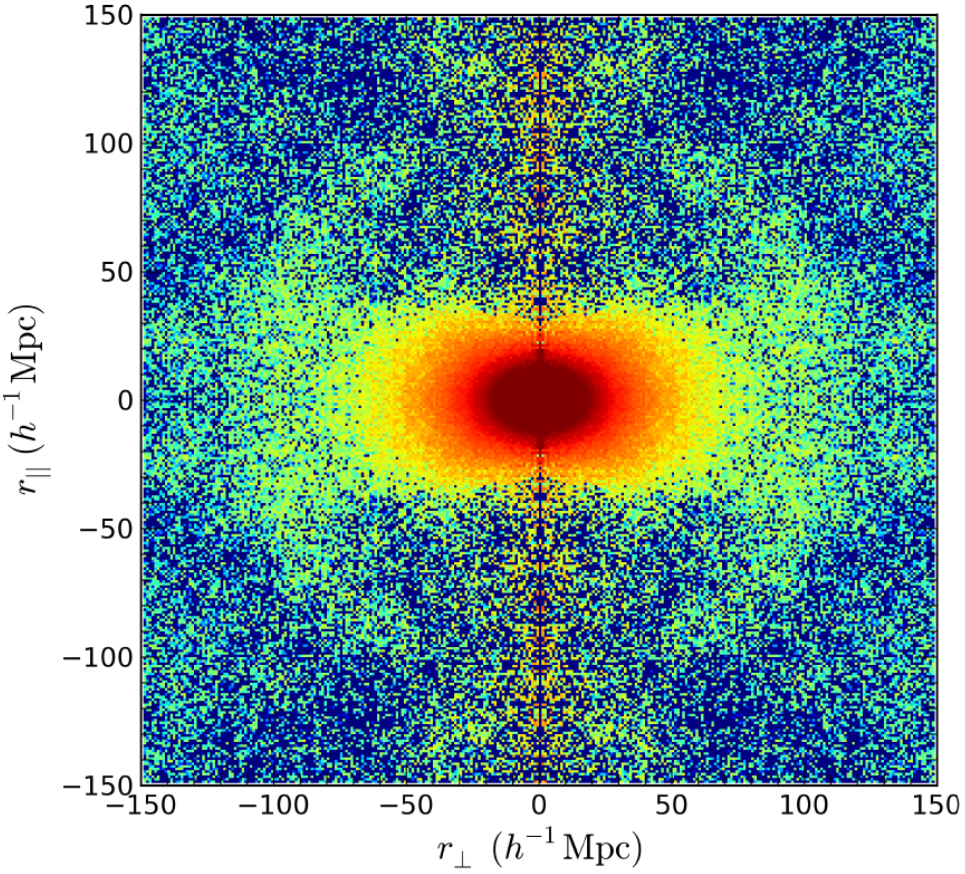


Figure 15: RSD

Angular correlation function:

$$w_g(\theta) = \int_0^\infty \frac{dl}{2\pi} l C_g(l) J_0(l\theta) \tag{122}$$

14.3 tSZ effect

Thermal Sunyaev-Zel'dovich (tSZ) effect: inverse-Compton scattering of high-temperature electrons in gas to CMB photons.

Gas pressure:

$$\mathcal{P}_{gas}(\mathbf{x}, \eta) = n_e(\mathbf{x}, \eta) T_e(\mathbf{x}, \eta) = \overline{n_e T_e}(\eta)[1 + \delta_{\mathcal{P}}(\mathbf{x}, \eta)] \tag{123}$$

tSZ term y is the integral over the kinetic energy density/pressure of the ionized gas along the line of sight:

$$y = \frac{\sigma_T}{m_e} \int n_e T_e a d\chi = \frac{\sigma_T}{m_e} \int \overline{n_e T_e}(\eta) a [1 + \delta_{\mathcal{P}}(\mathbf{x}, \eta)] d\chi \tag{124}$$

Distribution of CMB photons:

$$\begin{cases} f(x, y) \overset{y \ll 1}{=} [1 + y \frac{1}{x^2} \frac{\partial}{\partial x} (x^4 \frac{\partial}{\partial x})] \frac{1}{e^x - 1} \\ x = p/T \end{cases} \tag{125}$$

Radiation intensity at $T_0 = 2.73K$ in Figure 16

$$\begin{aligned} u_\nu(T, x, y) &= 4\pi I_\nu(T, x, y) \\ I_\nu(T, x, y) &= 4\pi\nu^3 f(x, y) \\ \nu &= p/(2\pi) = x \cdot T/(2\pi) \end{aligned} \tag{126}$$

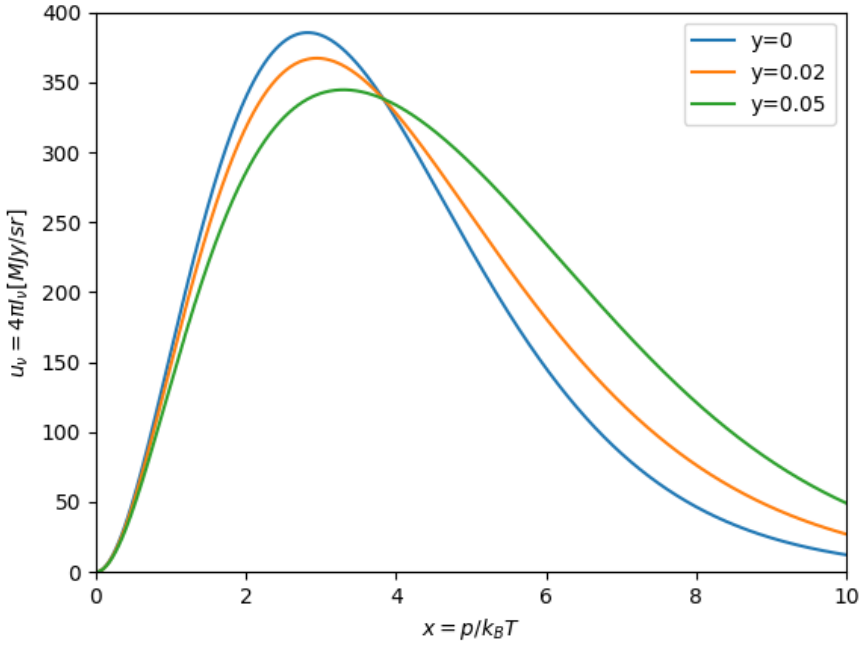


Figure 16: tSZ effect

Power spectrum of gas pressure perturbation:

$$\langle \delta_{\mathcal{P}}(\mathbf{k}_1)\delta_{\mathcal{P}}(\mathbf{k}_2)\rangle = (2\pi)^3 \delta_D^{(3)}(\mathbf{k}_1 + \mathbf{k}_2) P_{\mathcal{P}}(k_1) \tag{127}$$

Angular power spectrum of the tSZ y-distortion of the CMB:

$$C_y(l) = \left(\frac{\sigma_T}{m_e}\right)^2 \int_0^{\chi_*} \frac{d\chi}{\chi^2} [\overline{n_e T_e} a]^2 P_{\mathcal{P}}(k = \frac{l + 1/2}{\chi}, \eta(\chi)) \tag{128}$$

15 Perturbation Theory

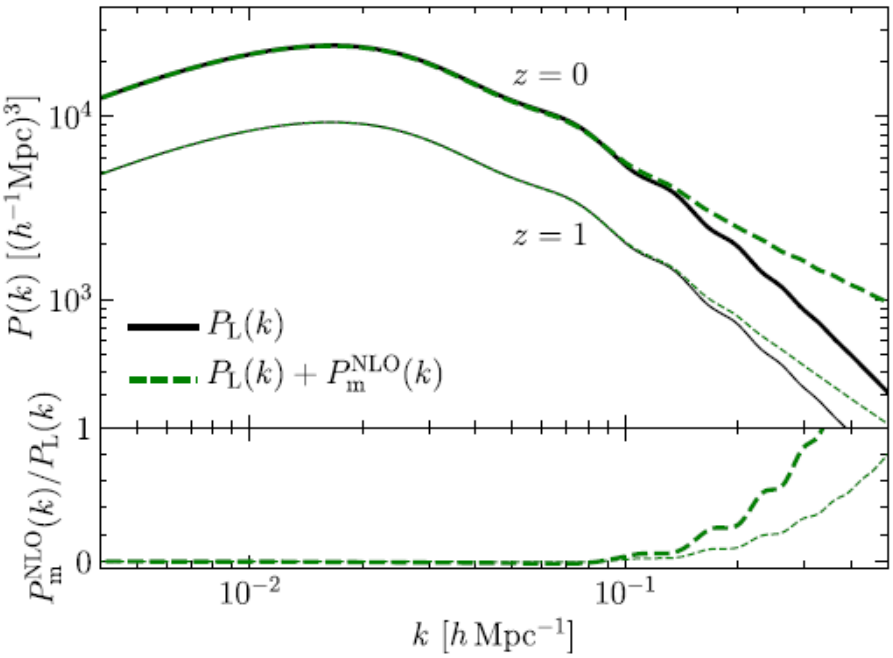


Figure 17: Power Spectrum in Perturbation Theory

Vlasov-Poisson System:

$$\begin{aligned} &\frac{\partial f_{\rm m}}{\partial t} + \frac{\partial f_{\rm m}}{\partial x^j}\frac{p^j}{ma} - \frac{\partial f_{\rm m}}{\partial p^j}\left[Hp^j + \frac{m}{a}\frac{\partial \Psi}{\partial x^j}\right] = 0, \\ &\nabla^2\Psi = 4\pi G a^2\left[\int \frac{d^3p}{(2\pi)^3} f_{\rm m}(x,p,t) - \rho_{\rm m}(t)\right]. \end{aligned} \tag{129}$$

Velocity divergence:

$$\theta_m = \partial_i u_m^i \tag{130}$$

From Vlasov-Poisson system, we get equations of perturbation theory:

$$\begin{aligned} {\delta_{\rm m}}' + \theta_{\rm m} &= -\delta_{\rm m}\theta_{\rm m} - u_{\rm m}^j\frac{\partial}{\partial x^j}\delta_{\rm m}, \\ {\theta_{\rm m}}' + aH\theta_{\rm m} + \nabla^2\Psi &= -u_{\rm m}^j\frac{\partial}{\partial x^j}\theta_{\rm m} - (\partial_i u_{\rm m}^j)(\partial_j u_{\rm m}^i), \\ \nabla^2\Psi &= \frac{3}{2}\Omega_{\rm m}(\eta)(aH)^2\delta_{\rm m}. \end{aligned} \tag{131}$$

Perturbation of matter density and velocity divergence:

$$\begin{aligned} \delta_{\rm m}(x,\eta) &= \delta^{(1)}(x,\eta) + \delta^{(2)}(x,\eta) + \cdots + \delta^{(n)}(x,\eta), \\ \theta_{\rm m}(x,\eta) &= \theta^{(1)}(x,\eta) + \theta^{(2)}(x,\eta) + \cdots + \theta^{(n)}(x,\eta), \end{aligned} \tag{132}$$

Solution of each order perturbation:

$$
\begin{aligned}
\delta^{(n)}(k,\eta) &= D_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times F_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n),\\
\theta^{(n)}(k,\eta) &= -aHfD_+^n(\eta)\left[\prod_{i=1}^n\int\frac{d^3k_i}{(2\pi)^3}\right](2\pi)^3\delta_{\rm D}^{(3)}\left(\mathbf{k}-\sum_{i=1}^n\mathbf{k}_i\right)\\
&\quad\times G_n(\mathbf{k}_1,\cdots,\mathbf{k}_n)\delta_0(\mathbf{k}_1)\cdots\delta_0(\mathbf{k}_n).
\end{aligned}
\tag{133}
$$

Where:

$$
\begin{aligned}
F_1 &= G_1 = 1\\
F_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{5}{7}+\frac{2}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
G_2(\mathbf{k}_1,\mathbf{k}_2) &= \frac{3}{7}+\frac{4}{7}\frac{(\mathbf{k}_1\cdot\mathbf{k}_2)^2}{k_1^2k_2^2}+\frac{1}{2}\widehat{\mathbf{k}}_1\cdot\widehat{\mathbf{k}}_2\left(\frac{k_1}{k_2}+\frac{k_2}{k_1}\right)\\
&\cdots
\end{aligned}
\tag{134}
$$

Power spectrum in Figure 17

$$
\begin{aligned}
&P(k,\eta) = P_{\rm L}(k,\eta)+P^{\rm NLO}(k,\eta)+\cdots,\\
&P^{\rm NLO}(k,\eta) = P^{(22)}(k,\eta)+2P^{(13)}(k,\eta),\\
&P^{(22)}(k,\eta) = 2\int\frac{d^3p}{(2\pi)^3}\Big[F_2(\mathbf{p},\mathbf{k}-\mathbf{p})\Big]^2P_{\rm L}(p,\eta)P_{\rm L}(|\mathbf{k}-\mathbf{p}|,\eta),\\
&P^{(13)}(k,\eta) = 3P_{\rm L}(k,\eta)\int\frac{d^3p}{(2\pi)^3}F_3(\mathbf{p},-\mathbf{p},\mathbf{k})P_{\rm L}(p,\eta).
\end{aligned}
\tag{135}
$$

Bispectrum:

$$
\begin{aligned}
\langle\delta_{\rm m}(\mathbf{k}_1,\eta)\delta_{\rm m}(\mathbf{k}_2,\eta)\delta_{\rm m}(\mathbf{k}_3,\eta)\rangle &= (2\pi)^3\delta_{\rm D}^{(3)}(\mathbf{k}_1+\mathbf{k}_2+\mathbf{k}_3)\\
&\quad\times\left[2F_2(\mathbf{k}_1,\mathbf{k}_2)P_{\rm L}(k_1,\eta)P_{\rm L}(k_2,\eta)+2\text{perm.}\right]
\end{aligned}
\tag{136}
$$

16 Spherical Collapse

16.1 Equations of spherical collapse

In Einstein-de Sitter Universe ($\Lambda = 0, k = 0$), and matter era. Omit pressure and baryon effects.

Newton's Gravitational Equation:

$$\ddot{r} = -\frac{GM}{r} \tag{137}$$

Energy conservation:

$$E = \frac{1}{2}\dot{r} - \frac{GM}{r} \tag{138}$$

Mass of perturbation M and energy E are constants.

16.2 Solution of spherical collapse

When $E < 0$, we get the solution:

$$\begin{aligned} r(\theta) &= \frac{GM}{2|E|}(1 - \cos\theta) \\ t(\theta) &= \frac{GM}{|2E|^{3/2}}(\theta - \sin\theta) \end{aligned} \tag{139}$$

16.3 Halo density perturbation

Halo density:

$$\rho_p(\theta) = \frac{M}{(4\pi/3)r^3} = \frac{6|E|^3}{\pi G^3 M^2}\frac{1}{(1-\cos\theta)^3} \tag{140}$$

Background density from Friedmann equation:

$$\bar{\rho}(\theta) = \frac{3H^2}{8\pi G} = \frac{1}{6\pi G t^2} = \frac{4|E|^3}{3\pi G^3 M^2}\frac{1}{(\theta - \sin\theta)^2} \tag{141}$$

Halo density perturbation:

$$1 + \delta_p(\theta) = \frac{\rho_p}{\bar{\rho}} = \frac{9}{2}\frac{(\theta - \sin\theta)^2}{(1-\cos\theta)^3} \propto \frac{t^2}{r^3} \tag{142}$$

Linear density perturbation ($\theta \ll 1$):

$$\delta_L(\theta) = \frac{3|E|}{10}\left(\frac{6}{GM}\right)^{2/3} t^{2/3} = \frac{3}{20}[6(\theta - \sin\theta)]^{2/3} \propto t^{2/3} \tag{143}$$

16.4 Perturbation on different stages

(1) Turn-around:

Time and radius:

$$\begin{aligned} \theta_{ta} &= \pi \\ t_{ta} = t(\theta_{ta}) &= \pi \cdot \frac{GM}{|2E|^{3/2}} \\ r_{ta} = r(\theta_{ta}) &= 2 \cdot \frac{GM}{2|E|} \end{aligned} \tag{144}$$

Perturbation:

$$\begin{aligned} \delta_{p,ta} &= \delta_p(\theta_{ta}) = 4.55 \\ \delta_{L,ta} &= \delta_L(\theta_{ta}) = 1.06 \end{aligned} \tag{145}$$

r(t) relation no longer holds at late collapse era, for pressure of baryons is large at late collapse era.

At late collapse era, compare with turn around perturbations and estimate perturbations from:

$$1 + \delta_p \propto \frac{t^2}{r^3}, \quad \delta_L \propto t^{2/3} \tag{146}$$

(2) Bounded:

The time when the halo is bounded.

Time and radius:

$$\begin{aligned} \theta_c &= \frac{3}{2}\pi \\ t_c = t(\theta_c) &\approx 2t_{ta} \\ r_c = r(\theta_c) &= \frac{1}{2}r_{ta} \end{aligned} \tag{147}$$

Perturbation:

$$\begin{aligned} 1 + \delta_{p,c} &= \left(\frac{t_c}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_c}\right)^3 (1 + \delta_{p,ta}) = 177.6 \\ \delta_{cr} = \delta_{L,c} &= \left(\frac{t_c}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 1.686 \end{aligned} \tag{148}$$

(3) Virialized:

The time when the halo is in virial equilibrium.

Time and radius:

$$\begin{aligned} \theta_{vir} &= 3\pi \\ t_{vir} = t(\theta_{vir}) &= 3t_{ta} \\ r_{vir} &= \frac{1}{2}r_{ta} \end{aligned} \tag{149}$$

Perturbation:

$$
\begin{aligned}
1 + \delta_{p,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^2 \left(\frac{r_{ta}}{r_{vir}}\right)^3 (1 + \delta_{p,ta}) = 400 \\
\delta_{L,vir} &= \left(\frac{t_{vir}}{t_{ta}}\right)^{2/3} \delta_{L,ta} = 2.20
\end{aligned}
\tag{150}
$$

17 The Halo Model

17.1 Halo Density Profile

Halo density Navarro, Frenk and White profile (NFW profile) in Figure 18

$$\rho_{\rm h}(r) = \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \tag{151}$$

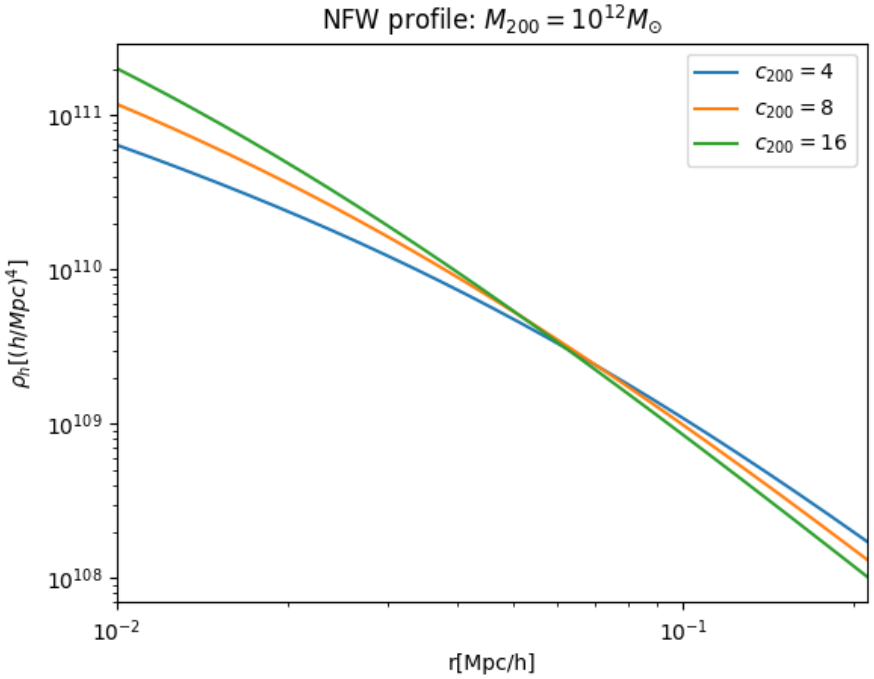


Figure 18: NFW Halo Density Profile $\rho_h(r)$

Concentration:

$$c_\Delta = R_\Delta / r_{sh} \tag{152}$$

Halo mass:

$$M_\Delta = M(< R_\Delta) = \frac{4\pi}{3} R_\Delta^3 \cdot \rho_{m0} \cdot \Delta \tag{153}$$

$$M_\Delta = \int_0^{R_\Delta} 4\pi r^2 \rho_h(r) dr \tag{154}$$

Usually set $\Delta = 200$.
For certain Δ,we can set $\rho_h(r)$ in terms of M_Δ and c_Δ:

$$
\begin{aligned}
\rho_{\rm h}(r) &= \frac{\rho_s}{(r/r_{sh})(1+r/r_{sh})^2} \\
R_\Delta &= \left(\frac{3M_\Delta}{4\pi\rho_{m0}\cdot\Delta}\right)^{1/3} \\
r_{sh} &= R_\Delta/c_\Delta \\
\rho_s &= \frac{M_\Delta}{4\pi r_{sh}^3}\left[\ln(c_\Delta+1)+\frac{1}{c_\Delta+1}-1\right]^{-1}
\end{aligned}
\tag{155}
$$

NFW halo density profile in k space:

$$
\begin{aligned}
\rho_h\left(k=\frac{x}{r_s},M\right) = \frac{M}{\mathcal{N}}[&\cos x\,[\mathrm{Ci}([c+1]x)-\mathrm{Ci}(x)] \\
&+\sin x\,[\mathrm{Si}([c+1]x)-\mathrm{Si}(x)]-\frac{\sin cx}{(c+1)x}]
\end{aligned}
\tag{156}
$$

$$
\begin{aligned}
\mathcal{N} &= \frac{1}{c+1}+\log(c+1)-1 \\
r_{sh} &= R_{200}/c, \quad c=8
\end{aligned}
\tag{157}
$$

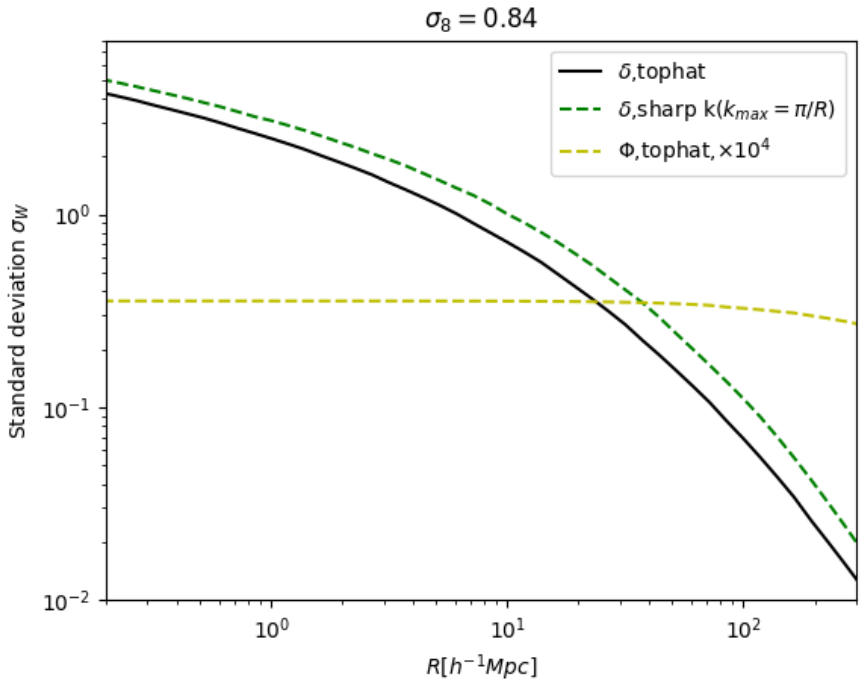


Figure 19: $\sigma_W(R)$

17.2 Halo Mass Function

Lagrangian radius (enclose mass of halo at the mean matter density):

$$
R_L(M) = 1.40h^{-1}\,\mathrm{Mpc}\left(\frac{M}{10^{12}h^{-1}M_\odot}\right)^{1/3} \tag{158}
$$

The top-hat window function describes a spherical perturbation with uniform density:

$$W_R(x) = \begin{cases} \frac{3}{4\pi R^3}, & 0 \le x < R \\ 0, & x \ge R \end{cases} \tag{159}$$

With FFT, we get top-hat window function in Fourier space:

$$W_R(k) = \frac{3}{(kR)^3}[\sin(kR) - (kR)\cos(kR)] \tag{160}$$

Correlation function in Figure 19

$$\begin{aligned} \sigma(M,z) &= \frac{D_+(z)}{D_+(z=0)}\sigma_W[R_L(M)] \\ \sigma_W^2(R) &= \frac{1}{2\pi^2}\int_0^\infty dk k^2 P_L(k)|W_R(k)|^2 \end{aligned} \tag{161}$$

And σ_8 :

$$\sigma_8 = \sigma_W(R = 8Mpc/h) \tag{162}$$

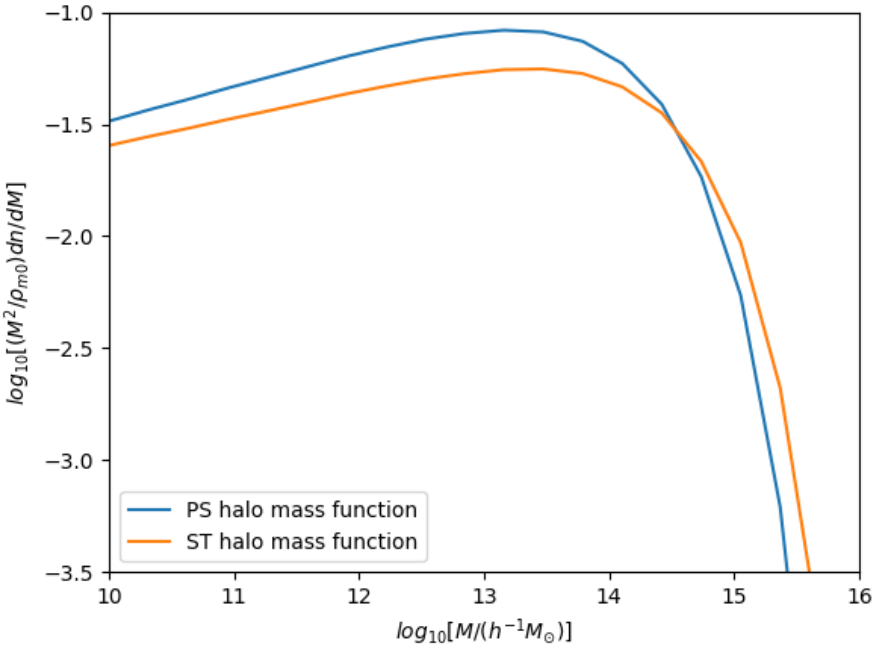


Figure 20: Halo Mass Function

Press-Schechter (PS) halo mass function:

$$\begin{aligned} \frac{dn(M,z)}{d\ln M} &= \frac{\rho_{\mathrm{m0}}}{M} f_{\mathrm{PS}}\left(\nu = \frac{\delta_{\mathrm{cr}}}{\sigma(M,z)}\right)\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right| \\ f_{\mathrm{PS}}(\nu) &= \sqrt{\frac{2}{\pi}}\nu e^{-\nu^2/2}, \quad \delta_{cr} = 1.686 \end{aligned} \tag{163}$$

Sheth-Tormen (ST) halo mass function:

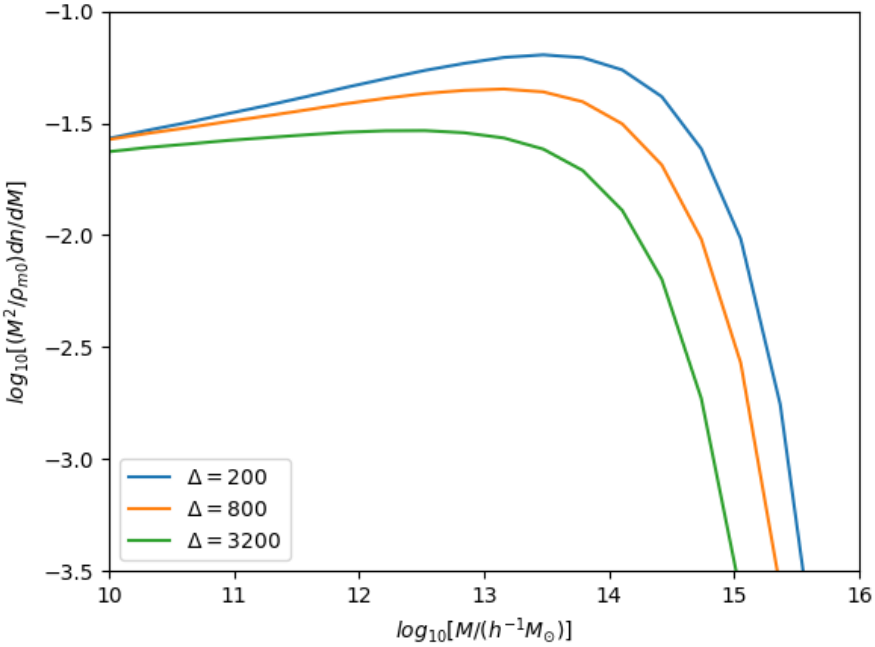


Figure 21: Tinker2008 Halo Mass Function

$$\begin{aligned}
\frac{dn(M,z)}{d\ln M} &= 0.322(1+\frac{1}{\nu'^{0.6}})\frac{\rho_{\rm m0}}{M}f_{\rm PS}\left(\nu'\right)\left|\frac{d\ln\sigma(M,z)}{d\ln M}\right| \\
f_{\rm PS}(\nu') &= \sqrt{\frac{2}{\pi}}\nu' e^{-\nu'^2/2}, \quad \delta_{cr} = 1.686 \\
\nu' &= \sqrt{0.707}\nu = \sqrt{0.707}\frac{\delta_{cr}}{\sigma(M,z)}
\end{aligned} \tag{164}$$

PS and ST halo mass function in Figure 20. Numerical halo mass function by Tinker2008 in Figure 21.

Number density of halos whose mass larger than M, in Figure 22

$$N(>M_1,z) = \int_{M_1}^{\infty} dn(M,z) = \int_{M_1}^{\infty}\frac{dn(M,z)}{d\ln M}d\ln M \tag{165}$$

17.3 Cluster of Halos

For PS halo mass function, halo - dark matter bias with normalization $P_{2h}(k) = P_L(k)$ for $k = 0.01h/Mpc$:

$$\begin{aligned}
\delta^{(1)}_{h,l}(\mathbf{x},t) &= b_1(M,z)\delta^{(1)}_l(\mathbf{x},t) \\
b_1^{PS}(M,z) &= 8.81\cdot\frac{\nu^2-1}{\delta_{cr}} \\
\nu &= \delta_{cr}/\sigma(M,z)
\end{aligned} \tag{166}$$

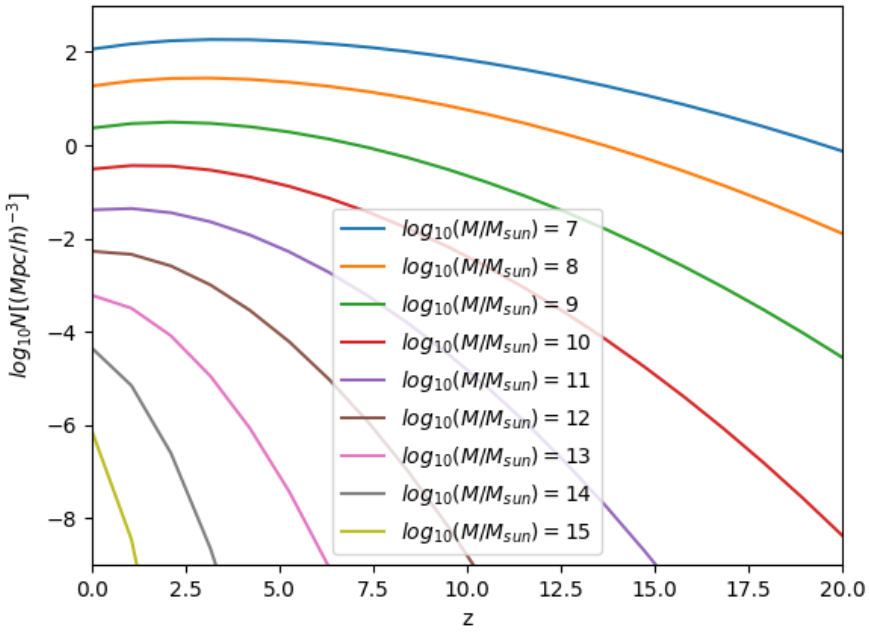


Figure 22: $N(>M, z)$

17.4 Power Spectrum in Halo Model

In observation, we get dark matter perturbation $\delta_{\rm m}^{\rm HM}$ from halo density profile $\rho_h(\mathbf{x}', M)$, halo mass function $dn/d\ln M$ and cluster of halos $\delta_h(\mathbf{x}', M)$.

Dark matter perturbation in halo model:

$$\delta_{\rm m}^{\rm HM}(x) = \int d\ln M \frac{1}{\rho_{\rm m0}} \frac{dn}{d\ln M} \int d^3x' \delta_h(\mathbf{x}', M) \rho_h(|\mathbf{x} - \mathbf{x}'|, M) \tag{167}$$

Power spectrum of halo model (NFW density profile + PS halo mass function + linear halo bias) in Figure 23

$$\begin{aligned} P^{\rm HM}(k, a) &= P_{\rm 2h}(k, a) + P_{\rm 1h}(k, a), \\ P_{\rm 2h}(k, a) &= \left[\frac{1}{\rho_{\rm m0}} \int d\ln M \frac{dn}{d\ln M} b_1(M, a) \rho_h(k, M)\right]^2 P_{\rm L}(k, a), \\ P_{\rm 1h}(k, a) &= \frac{1}{\rho_{\rm m0}^2} \int d\ln M \frac{dn}{d\ln M} [\rho_h(k, M)]^2. \end{aligned} \tag{168}$$

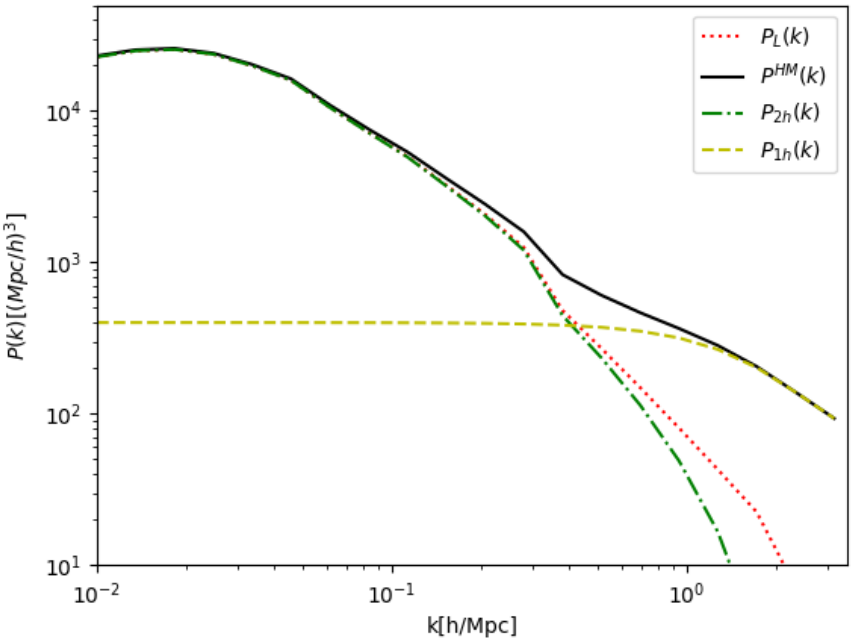


Figure 23: $P^{\mathrm{HM}}(k, a = 1)$

18 Galaxy Clusters

18.1 Mass-Temperature Relation

Kinetic energy:

$$E_k = \frac{1}{2} M \langle v^2 \rangle \tag{169}$$

Potential energy:

$$E_p = -\frac{3}{5} \frac{GM^2}{R_{vir}} \tag{170}$$

Halo in virial equilibrium:

$$E_k = -\frac{1}{2} E_p \tag{171}$$

Ideal gas in thermodynamic equilibrium:

$$\frac{1}{2} m_p \langle v^2 \rangle = \frac{3}{2} kT \tag{172}$$

Mass-Temperature relation of halo:

$$M = 1.38 \times 10^{14} h^{-1} M_\odot \left(\frac{T}{\mathrm{keV}} \right)^{3/2} \left(\frac{\Delta_{\mathrm{vir}}}{200} \right)^{-1/2} \tag{173}$$

Usually we set overdensity of halo $\triangle_{vir} = 200$

18.2 Power Spectrum with top-hat window function

Power spectrum of galaxy cluster / halo whose mass is M and density perturbation beyond δ_{cr} with top-hat window function at $z = 0$:

$$\begin{aligned} P_{\mathrm{g,thr}}(k, M) &= (b_1^{\mathrm{thr}})^2 P_{\mathrm{L}}(k) W_R^2(k) \\ &+ \frac{1}{2}(b_2^{\mathrm{thr}})^2 \int \frac{d^3p}{(2\pi)^3} P_{\mathrm{L}}(p) W_R^2(p) P_{\mathrm{L}}(|\mathbf{k}-\mathbf{p}|) W_R^2(|\mathbf{k}-\mathbf{p}|) \\ &+ \cdots \end{aligned} \tag{174}$$

Where:

$$\begin{aligned} b_1^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2}}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma(M)} \overset{\nu \gg 1}{\approx} \frac{\nu}{\sigma(M)} \\ b_2^{\mathrm{thr}} &= \sqrt{\frac{2}{\pi}} \frac{e^{-\nu^2/2} \cdot \nu}{\mathrm{erfc}[\nu/\sqrt{2}]\sigma^2(M)} \overset{\nu \gg 1}{\approx} \frac{\nu^2}{\sigma^2(M)} \\ \nu &= \delta_{cr}/\sigma(M) \\ R &= R_L(M) \end{aligned} \tag{175}$$

19 Gravitational Lensing

19.1 Angle Deflection

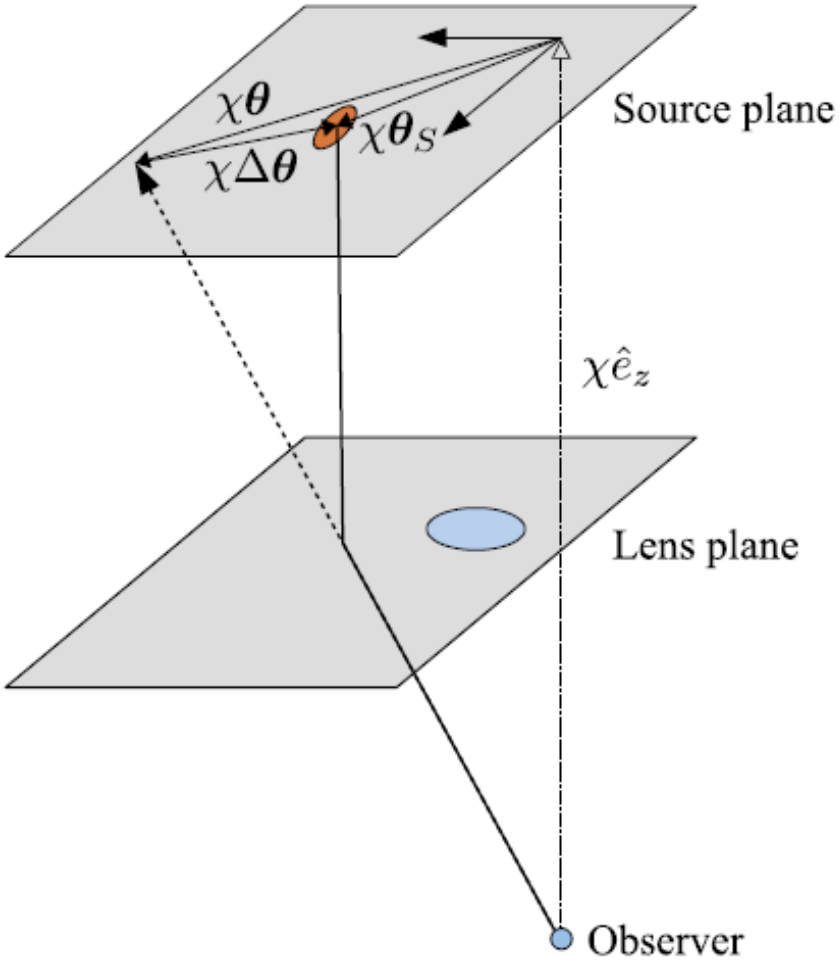


Figure 24: Gravitational Lensing

Gravitational lensing in Figure 24
Photon geodesics:

$$\frac{d\hat{p}}{d\chi} = -2(\delta^{ik} - \hat{p}^i\hat{p}^k)\Phi_{,k} \tag{176}$$

Lensing potential:

$$\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi(\theta,\chi')(1-\frac{\chi'}{\chi}) \tag{177}$$

Deflection angle:

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial\theta^i}\phi_L(\theta) \tag{178}$$

Distortion tensor:

$$-\Psi_{ij} = -\frac{\partial^2}{\partial\theta^i\partial\theta^j}\phi_L(\theta) = \kappa\delta_{ij} + \begin{pmatrix}\gamma_1 & \gamma_2\\ \gamma_2 & -\gamma_1\end{pmatrix} \tag{179}$$

Convergence:κ. Shear:γ_1, γ_2.
Angle deflection matrix:

$$A_{ij} = \frac{\partial \theta_s^i}{\partial \theta^i} = \delta_{ij} + \Psi_{ij} \tag{180}$$

19.2 Lensing of Point Mass

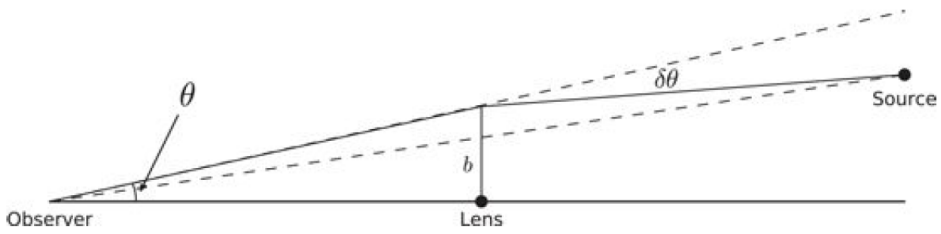


Figure 25: Point mass lensing to a very distant source

Deflection angle of lensing:

$$\Delta\theta^i = \theta_s^i - \theta^i = \frac{\partial}{\partial \theta^i}\phi_L(\theta) = 2\int_0^\chi \frac{d\chi'}{\chi'}\Phi_{,i}(\theta, \chi')(1 - \frac{\chi'}{\chi}) \tag{181}$$

Point mass lensing to a very distant source, in Figure 25 from "Gravitational Lensing" of Dodelson. [1]
For point mass at $(0, \chi_{lens})$,from Newton's Law of Gravity,

$$\Phi(y = \chi\theta, \chi) = \frac{GM}{\sqrt{y^2 + (\chi - \chi_{lens})^2}} \tag{182}$$

Source is very distant, $\chi_S \to \infty$, and $b \ll \chi_{lens}$,

$$\Delta\theta^i = 2\int_0^\infty \frac{d\chi'}{\chi'}\frac{\partial\Phi}{\partial\theta^i}\bigg|_{\chi\theta=b} = 2\int_0^\infty d\chi'\frac{\partial\Phi}{\partial y}\bigg|_{y=b} \tag{183}$$

We get:

$$\Delta\theta = -\frac{4GM}{b} \tag{184}$$

This result is verified in Eddington's eclipse expedition.

19.3 Galaxy Shape

Galaxy shape tensor:

$$q_{ij} = \frac{1}{2}q\delta_{ij} + \frac{1}{2}q\begin{pmatrix}\epsilon_1 & \epsilon_2 \\ \epsilon_2 & -\epsilon_1\end{pmatrix} \tag{185}$$

$$q_{ij} = \langle\theta_i\theta_j\rangle_{I_{obs}} = \frac{1}{F}\int d^2\theta I_{obs}(\theta)\theta_i\theta_j \tag{186}$$

Total flux:

$$F = \int d^2\theta I_{obs}(\theta) \tag{187}$$

Polarization tensor:

$$I_{ij} = I\delta_{ij} + \begin{pmatrix} Q & U \\ U & -Q \end{pmatrix} \tag{188}$$

From conservation of intensity:

$$I_{obs}(\theta) = I_{true}(\theta_s) \tag{189}$$

We get the relation of observed ellipticity of galaxy with true ellipticity , when $\kappa, \gamma_1, \gamma_2 \ll 1$:

$$\begin{aligned} \epsilon_1 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= (1 - 2[\epsilon_1^{true}\gamma_1 + \epsilon_2^{true}\gamma_2])\epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{190}$$

If true ellipticity is small $\epsilon_1^{true}, \epsilon_2^{true} \ll 1$:

$$\begin{aligned} \epsilon_1 &= \epsilon_1^{true} + 2\gamma_1 \\ \epsilon_2 &= \epsilon_2^{true} + 2\gamma_2 \end{aligned} \tag{191}$$

Intrinsic alignment:

$$q_{ij}^{IA} = -c_1(\eta)\chi^2\Phi_{,ij}(\mathbf{x}, \eta) \tag{192}$$

19.4 Lensing/Shear Angular Power Spectrum

In Fourier space:

$$\mathbf{l} = l(\cos\theta_l, \sin\theta_l) \tag{193}$$

Change shear γ_1, γ_2 to rotation-invariant E,B:

$$\begin{pmatrix} \gamma_1(\mathbf{l}) \\ \gamma_2(\mathbf{l}) \end{pmatrix} = \begin{pmatrix} \cos(2\phi_l) & -\sin(2\phi_l) \\ \sin(2\phi_l) & \cos(2\phi_l) \end{pmatrix} \begin{pmatrix} E(\mathbf{l}) \\ B(\mathbf{l}) \end{pmatrix} \tag{194}$$

In Fourier space:

$$\begin{aligned} E(\mathbf{l}) &= (\frac{l^i l^j}{l^2} - \frac{1}{2}\delta_{ij})(-\Psi_{ij}) \\ B(\mathbf{l}) &= \epsilon_{ik}\frac{l^j l^k}{l^2}(-\Psi_{ij}) \end{aligned} \tag{195}$$

We get:

$$\begin{aligned} E(\mathbf{l}) = \kappa(\mathbf{l}) &= \frac{1}{2}l^2\phi_L(\mathbf{l}) \\ B(\mathbf{l}) &= 0 \end{aligned} \tag{196}$$

Lensing/Shear angular power spectrum of galaxies in Figure 26

$$
\begin{aligned}
C_{EE}(l) = C_{\kappa\kappa}(l) &= \frac{1}{4} l^4 C_{\phi_L \phi_L}(l) \\
&= (\frac{3}{2}\Omega_{m0} H_0^2)^2 \int_0^\infty d\chi a^{-2}(\chi) g_L^2(\chi) P(k = \frac{l}{\chi}, \eta = \eta_0 - \chi) \\
g_L(\chi) &= \int_\chi^\infty d\chi' (1 - \frac{\chi}{\chi'}) W(\chi') \\
W(\chi) &= \frac{1}{N_g} \frac{dN_g}{d\chi}
\end{aligned}
\tag{197}
$$

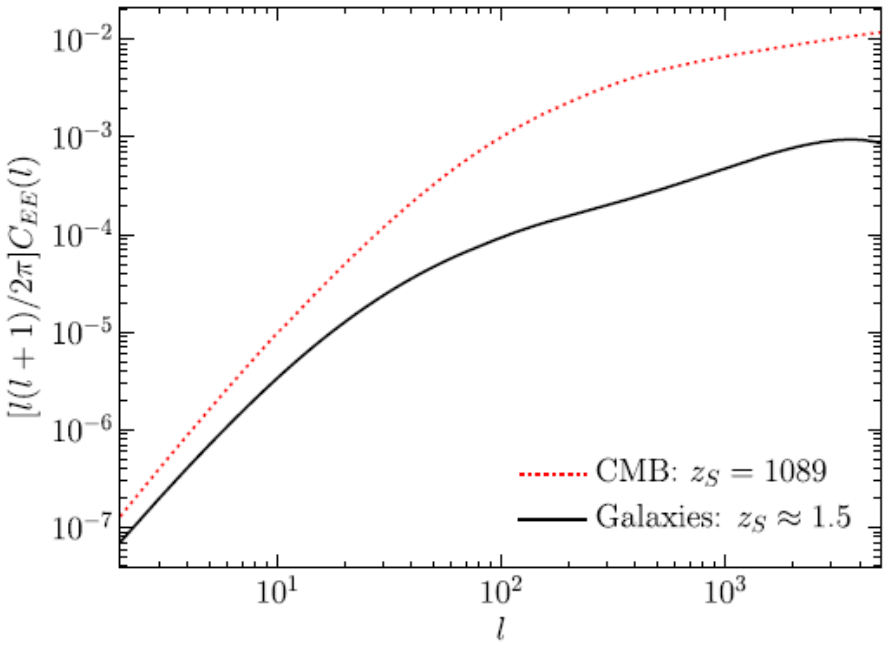


Figure 26: E-mode shear power spectrum

Change shear γ_1, γ_2 to rotation-invariant tangent shear γ_t and cross shear $\gamma_\times$.

Shear correlation functions:

$$
\begin{aligned}
\xi_{+,-}(\theta) &= \langle \gamma_t(\mathbf{0}) \gamma_t(\theta) \rangle \pm \langle \gamma_\times(\mathbf{0}) \gamma_\times(\theta) \rangle \\
&= \langle \gamma_1(\mathbf{0}) \gamma_1(\theta) \rangle \pm \langle \gamma_2(\mathbf{0}) \gamma_2(\theta) \rangle \\
&= \int \frac{l dl}{2\pi} J_{0,4}(l\theta) C_{EE}(l)
\end{aligned}
\tag{198}
$$

19.5 Shear Cross Angular Power Spectrum

Projected galaxy density:

$$
\Delta_g(\theta) = \int_0^\infty d\chi W_g(\chi) \delta_g(\theta, \eta = \eta_0 - \chi) \tag{199}
$$

On large scale, galaxy-matter cross power spectrum satisfies:

$$
P_{gm}(M, k, z) = b_1(M, z) P(k, z), \quad (k \ll 0.1h/Mpc) \tag{200}
$$

Shear cross angular power spectrum:

$$C_{gE}(l) = (\frac{3}{2}\Omega_{m0}H_0^2)\int_0^\infty \frac{d\chi}{\chi}W_g(\chi)a^{-1}(\chi)g_L(\chi)P_{gm}(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{201}$$

Shear cross correlation functions

$$\begin{aligned}\xi_{g,t}(\theta) = \langle\Delta_g(\mathbf{0})\gamma_t(\theta)\rangle = -\int \frac{ldl}{2\pi}J_2(l\theta)C_{gE}(l)\\ \xi_{g,\times} = \langle\Delta_g(\mathbf{0})\gamma_\times(\theta)\rangle = 0\end{aligned} \tag{202}$$

19.6 CMB Lensing

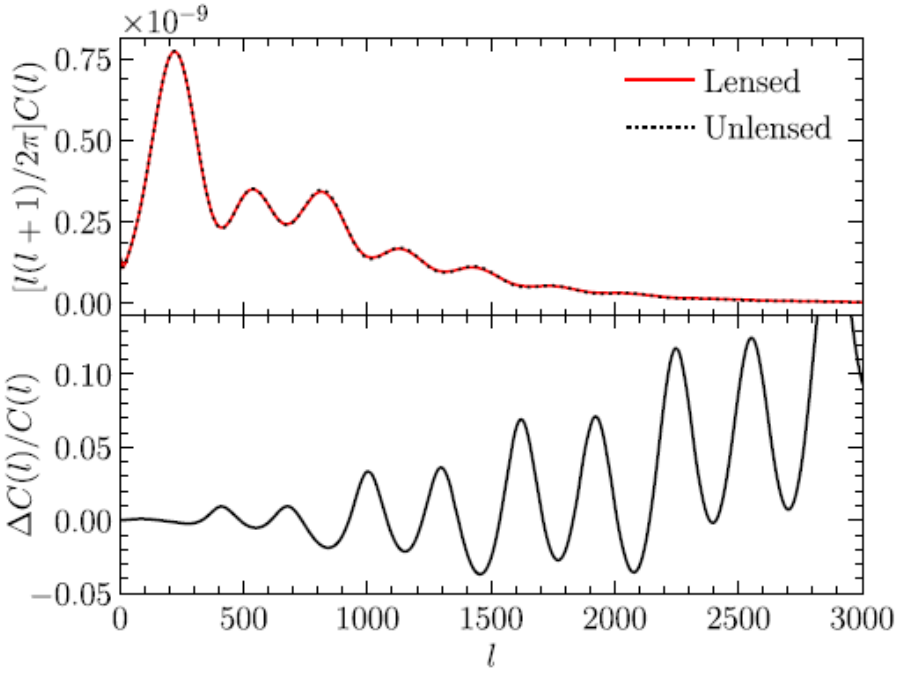


Figure 27: Effect of lensing on the CMB temperature power spectrum

From angle deflection due to gravitational lensing:

$$T_{obs}(\theta) = T(\theta_s) \tag{203}$$

Lensing angular power spectrum of CMB:

$$C_{\phi_L\phi_L}(l) = \frac{4}{l^4}(\frac{3}{2}\Omega_{m0}H_0^2)^2\int_0^{\chi_*} d\chi a^{-2}(\chi)(1-\frac{\chi}{\chi_*})^2 P(k=\frac{l}{\chi},\eta=\eta_0-\chi) \tag{204}$$

Angular power spectrum of CMB lensing in Figure 27.

$$\begin{aligned}C^{obs}(l) &= (1-l^2/l_{lens}^2)C(l) + C^{(22)}(l)\\ l_{lens}^{-2} &= \frac{1}{4\pi}\int_0^\infty dl_1 l_1^3 C_{\phi_L\phi_L}(l_1)\\ C^{(22)}(l) &= \frac{1}{(2\pi)^2}\int_0^{2\pi}d\theta\int_0^\infty dl_1 l_1(ll_1\cos\theta - l_1^2)^2 C_{\phi_L\phi_L}(l_1)C(\sqrt{l^2+l_1^2-2ll_1\cos\theta})\end{aligned} \tag{205}$$

20 Analysis

20.1 Overview: From Raw Data to Parameter Constraints

The process in Figure 28

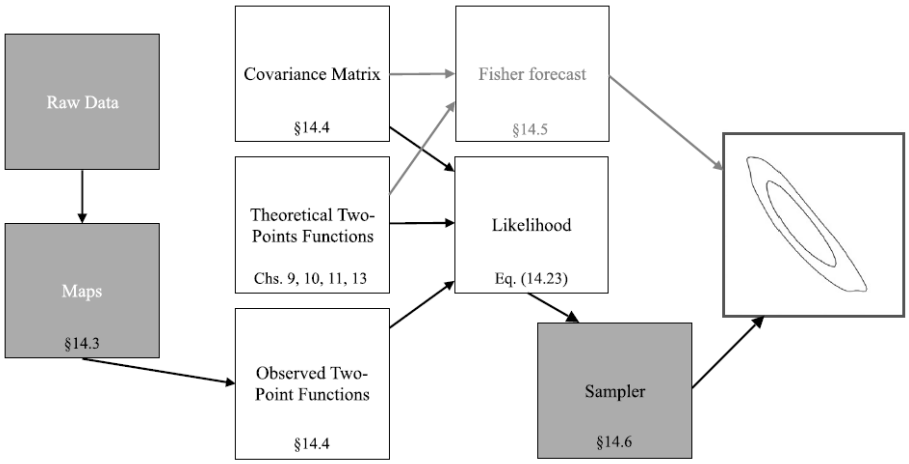


Figure 28: Overview: From Raw Data to Parameter Constraints

20.2 The Likelihood Function

Bayes' Theorem:

$$P(w, \sigma_w|\{d_i\}) = \frac{P(\{d_i\}|w, \sigma_w)P(w, \sigma_w)}{P(\{d_i\})} \tag{206}$$

$$P(w, \sigma_w|\{d_i\}) \propto \mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) P_{\rm prior}(w, \sigma_w) \tag{207}$$

Marginalized posterior:

$$P(w|\{d_i\}) = \int_0^\infty d\sigma_w P(w, \sigma_w|\{d_i\}) \tag{208}$$

The likelihood function for Gaussian Distribution:

$$\mathcal{L}\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) = P\left(\{d_i\}_{i=1}^m \middle| w, \sigma_w\right) = \frac{1}{(2\pi\sigma_w^2)^{m/2}} \exp\left\{-\frac{\sum_{i=1}^m (d_i - w)^2}{2\sigma_w^2}\right\} \tag{209}$$

To maximum the likelihood function:

$$\frac{\partial \mathcal{L}}{\partial w} = 0, \quad \frac{\partial \mathcal{L}}{\partial \sigma_w^2} = 0 \tag{210}$$

We get:

$$\hat{w} = \frac{1}{m}\sum_{i=1}^m d_i, \quad \widehat{\sigma_w^2} = \frac{1}{m}\sum_{i=1}^m (d_i - w)^2 \tag{211}$$

Expand the likelihood function:

$$\begin{aligned}\ln \mathcal{L}(w) &= \ln \mathcal{L}(\hat{w}) + \left.\frac{\partial \ln \mathcal{L}}{\partial w}\right|_{w=\hat{w}} (w-\hat{w}) + \frac{1}{2}\left.\frac{\partial^2 \ln \mathcal{L}}{\partial w^2}\right|_{w=\hat{w}} (w-\hat{w})^2 \\ &= \ln \mathcal{L}(\hat{w}) - \frac{m}{2\sigma_w^2}(w-\hat{w})^2,\end{aligned} \tag{212}$$

We get:

$$\mathrm{Var}[\hat{w}] = \left\langle (\hat{w}-w)^2 \right\rangle = -\big[\frac{\partial^2 \ln \mathcal{L}}{\partial w^2}\big]^{-1} = \frac{\sigma_w^2}{m} \tag{213}$$

$$\mathrm{Var}(\sigma_w^2) \equiv \left\langle \left(\widehat{\sigma_w^2} - \sigma_w^2\right)^2 \right\rangle = \frac{2}{m}\sigma_w^4 \tag{214}$$

20.3 Example: CMB Power Spectrum

For a CMB experiment, we infer parameters $\{\lambda_\alpha\}$ from observed temperature anisotropy $\hat{C}(l)$

CMB temperature anisotropy we observe:

$$\Delta = \frac{T - T_0}{T_0} \tag{215}$$

Considering beam pattern $B(\hat{\mathbf{n}}, \hat{\mathbf{n}}')$ and noise $\eta(\hat{\mathbf{n}})$

$$\Delta(\hat{\mathbf{n}}) = \int d\Omega' \Theta(\hat{\mathbf{n}}') B(\hat{\mathbf{n}}, \hat{\mathbf{n}}') + \eta(\hat{\mathbf{n}}) \tag{216}$$

In spherical harmonics:

$$a_{lm}^{\mathrm{obs}} = \int d\Omega Y_{lm}^*(\hat{\mathbf{n}}) \Delta(\hat{\mathbf{n}}) = \sum_{l'm'} B_{lm,l'm'} a_{l'm'} + \eta_{lm} \tag{217}$$

If the beam is isotropic $B(\hat{\mathbf{n}}, \hat{\mathbf{n}'}) = B(\hat{\mathbf{n}} \cdot \hat{\mathbf{n}'})$:

$$a_{lm}^{obs} = a_{lm} B_l + \eta_{lm} \tag{218}$$

We assume noise has mean zero with power spectrum:

$$\langle \eta_{lm} \eta_{l'm'}^* \rangle = N(l) \delta_{ll'} \delta_{mm'} \tag{219}$$

From:

$$a_{lm} | C(l) \sim N(\mu = 0, \sigma^2 = C(l)) \tag{220}$$

We know:

$$a_{lm}^{obs} | C(l) \sim N(\mu = 0, \sigma^2 = C(l) B_l^2 + N(l)) \tag{221}$$

The likelihood function:

$$\mathcal{L} \equiv P\left(\{a_{lm}^{\mathrm{obs}}\} | C(l)\right) = \left(2\pi \left[C(l)B_l^2 + N(l)\right]\right)^{-(2l+1)/2} \exp\left\{ -\frac{1}{2} \sum_{m=-l}^{l} \frac{\left|a_{lm}^{\mathrm{obs}}\right|^2}{C(l)B_l^2 + N(l)} \right\}. \tag{222}$$

To maximize $\mathcal{L}$, $d\ln\mathcal{L}/dC(l) = 0$, we get:

$$\hat{C}(l) = B_l^{-2}\left(\frac{1}{2l+1}\sum_{m=-l}^{l}\left|a_{lm}^{\rm obs}\right|^2 - N(l)\right) \tag{223}$$

$$\mathrm{Var}\left[\hat{C}(l)\right] = \left\langle\hat{C}(l)^2\right\rangle - C(l)^2 = \frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2 \tag{224}$$

Different l modes are independent:

$$\mathrm{Cov}_{ll'} = \mathrm{Var}\left[\hat{C}(l)\right]\delta_{ll'} = \frac{2}{2l+1}[C(l)+N(l)B_l^{-2}]^2\delta_{ll'} \tag{225}$$

For a Gaussian beam pattern:

$$B_l = e^{-l^2\theta_{beam}^2/2} \tag{226}$$

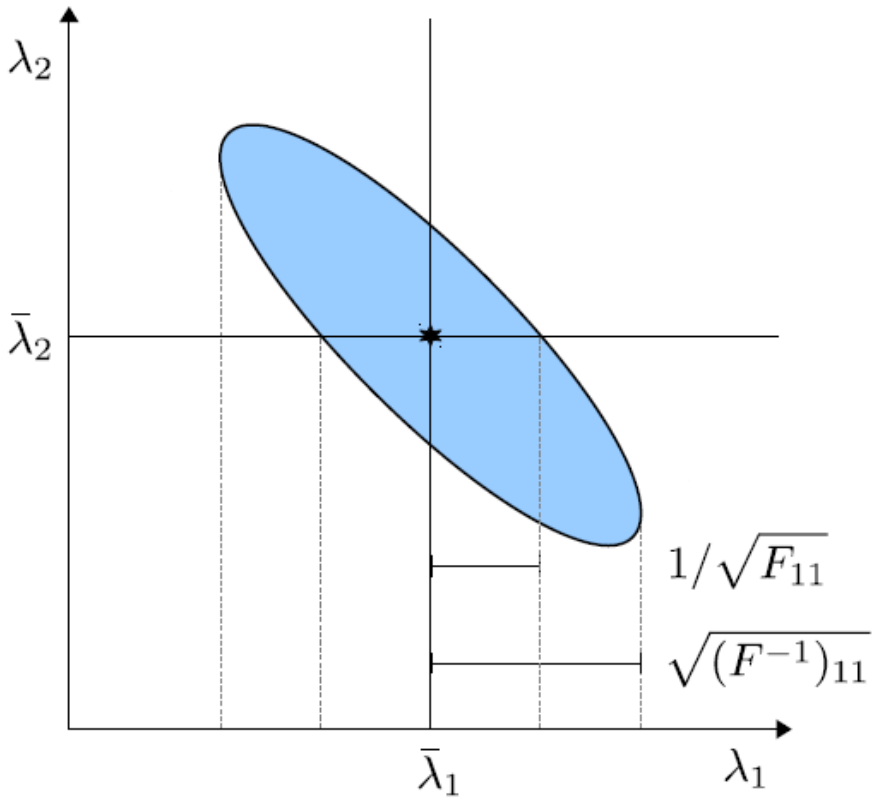


Figure 29: Error Ellipse from Fisher Matrix

The likelihood function for CMB anisotropies:

$$\ln\mathcal{L}(\{\lambda_\gamma\}) = -\frac{1}{2}\sum_{ll'}\left(\hat{C}(l) - C^{\rm theory}(l,\{\lambda_\gamma\})\right)\left(\mathrm{Cov}^{-1}\right)_{ll'}\left(\hat{C}(l') - C^{\rm theory}(l',\{\lambda_\gamma\})\right) \tag{227}$$

So,

$$\ln \mathcal{L}(\{\lambda_\gamma\}) = -\frac{\chi^2(\{\lambda_\gamma\})}{2} \tag{228}$$

$$\chi^2(\{\lambda_\gamma\}) = \sum_l \frac{[\hat{C}(l) - C^{\rm theory}(l, \{\lambda_\gamma\})]^2}{{\rm Var}\Big[\hat{C}(l)\Big]} \tag{229}$$

To maximize χ^2, $d\chi^2/d\lambda_\gamma = 0$, we expand χ^2 to 2nd order:

$$\chi^2(\{\lambda_\gamma\}) = \chi^2(\{\bar{\lambda}_\gamma\}) + \sum_{\alpha\beta} F_{\alpha\beta}(\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \tag{230}$$

The Fisher Matrix is defined as:

$$\begin{aligned} F_{\alpha\beta} &= \frac{1}{2}\left\langle \frac{\partial^2 \chi^2}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \\ &\equiv -\left\langle \frac{\partial^2 \ln \mathcal{L}}{\partial\lambda_\alpha \partial\lambda_\beta} \right\rangle \Bigg|_{\{\lambda_\gamma\}=\{\bar{\lambda}_\gamma\}} \end{aligned} \tag{231}$$

$$(F^{-1})_{\alpha\beta} = \langle (\lambda_\alpha - \bar{\lambda}_\alpha)(\lambda_\beta - \bar{\lambda}_\beta) \rangle \tag{232}$$

For CMB anisotropies:

$$F_{\alpha\beta} = \sum_l \frac{1}{{\rm Var}\Big[\hat{C}(l)\Big]} \frac{\partial C^{\rm theory}(l, \{\bar{\lambda}_\gamma\})}{\partial\bar{\lambda}_\alpha} \frac{\partial C^{\rm theory}(l, \{\bar{\lambda}_\gamma\})}{\partial\bar{\lambda}_\beta} \tag{233}$$

From Fisher Matrix, we can draw error ellipse for parameters in Figure 29.
Joint posterior for λ_1 and λ_2 :

$$P(\lambda_1, \lambda_2) \propto \exp\left\{-\frac{1}{2}\lambda_\alpha F_{\alpha\beta}\lambda_\beta\right\} \tag{234}$$

So,if λ_2 is known, error of λ_1 is : $\sqrt{\mathbf{Var}(\lambda_1)} = 1/\sqrt{F_{11}}$
Varying λ_2, posterior for λ_1 :

$$\begin{aligned} P(\lambda_1) &= \int d\lambda_2 P(\lambda_1, \lambda_2) \\ &\propto \exp\left\{-\frac{\lambda_1^2}{2}\left(\frac{F_{11}F_{22} - F_{12}F_{21}}{F_{22}}\right)\right\} = \exp\left\{-\frac{\lambda_1^2}{2(F^{-1})_{11}}\right\} \end{aligned} \tag{235}$$

So,if λ_2 is unknown, error of λ_1 is : $\sqrt{\mathbf{Var}(\lambda_1)} = \sqrt{(F^{-1})_{11}}$

20.4 Example: Galaxy Power Spectrum

1. Galaxy Covariance Matrix
The number perturbation of galaxy in pixel i:

$$\begin{aligned} \Delta_i &= \int d^3x \psi_i(\mathbf{x})[\frac{n(\mathbf{x}) - \bar{n}(\mathbf{x})}{\bar{n}(\mathbf{x})}] \\ \psi_i(\mathbf{x}) &= \frac{1}{V} e^{i\mathbf{k}_i\cdot\mathbf{x}} \end{aligned} \tag{236}$$

Covariance Matrix:

$$\langle \Delta_i \Delta_j^* \rangle = (C_S)_{ij} + (C_N)_{ij} \tag{237}$$

Shot noise from Poisson Distribution particles $n_{shot} \sim Pois(\bar{n})$:

$$P(n) = \frac{(\bar{n})^n e^{-\bar{n}}}{n!}, \quad n \in \{0, 1, 2, \dots\} \tag{238}$$

Covariance Matrix:

$$\begin{gathered}(C_N)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{PoissonNoise} = \delta_{ij} \frac{1}{N} \\ N = nV\end{gathered} \tag{239}$$

$$\begin{gathered}(C_S)_{ij} = \langle \Delta_i \Delta_j^* \rangle \Big|_{NoNoise} = \int_0^\infty \frac{dk}{k} [\frac{k^3 P(k)}{2\pi^2}] W_{ij}(k) \\ W_{ij}(k) = \delta_{ij} \frac{9}{2kk_iR^2} \int_{|k-k_i|}^{(k+k_i)R} \frac{dy}{y} j_1^2(y)\end{gathered} \tag{240}$$

At $k_iR \gg 1$:

$$(C_S)_{ii} = \frac{1}{V} P(k_i) \tag{241}$$

$$\frac{(C_S)_{ii}}{(C_N)_{ii}} = P(k_i)\bar{n} \tag{242}$$

2.Galaxy Power Spectrum

In x space, $V = L^3$.

In k space, we have: number of grids in an axis K_{grid}, edge length of a grid $k_F = 2\pi/L$, total length in half axis $k_{N_y} = K_{grid}k_F/2$.

FFT of perturbation:

$$\begin{gathered}\delta_g(\mathbf{k}) = L^{3/2} \sum_i^{K_{grid}^3} \delta_g(\mathbf{x}_i) e^{-i\mathbf{k}\cdot\mathbf{x}_i} \\ \delta_g(\mathbf{x}) = \frac{1}{K_{grid}^3 L^{3/2}} \sum_{\mathbf{k}}^{k_{N_y}} \delta_g(\mathbf{k}_i) e^{i\mathbf{k}_i\cdot\mathbf{x}}\end{gathered} \tag{243}$$

The number of modes that each volume α contains, for half of Fourier modes ($k_\alpha - \Delta k/2 \leq |\mathbf{k}| \leq k_\alpha + \Delta k/2$):

$$m_{k,\alpha} = \frac{1}{2} \frac{4\pi k_\alpha^2 \Delta k}{k_F^3} = \frac{1}{4\pi^2} V k_\alpha^2 \Delta k \tag{244}$$

Omit beam pattern effects on galaxy $B_l = 1$, we estimate galaxy power spectrum as:

$$\hat{P}_g(k_\alpha) = \frac{1}{m_{k,\alpha}} \sum_{\mathbf{k}}^{||\mathbf{k}|-k_\alpha|<\Delta k/2} |\delta_g(\mathbf{k})|^2 - P_N \tag{245}$$

δ_g is Gaussian, we get standard deviation:

$$\sqrt{\mathbf{Var}[\hat{P}_g(k_\alpha)]} = \sqrt{\langle \hat{P}_g(k_\alpha)^2 \rangle - \langle \hat{P}_g(k_\alpha) \rangle^2} = \sqrt{\frac{2}{m_{k,\alpha}}} [P_g(k_\alpha) + P_N] \tag{246}$$

Covariance matrix:

$$\mathbf{Cov}_{\alpha\beta} = \sqrt{\langle \hat{P}_g(k_\alpha) \hat{P}_g(k_\beta) \rangle - \langle \hat{P}_g(k_\alpha) \rangle \langle \hat{P}_g(k_\beta) \rangle} = \frac{2}{m_{k,\alpha}} [P_g(k_\alpha) + P_N]^2 \delta_{\alpha\beta} \tag{247}$$

For discrete Fourier-space field:

$$\langle \delta_g(\mathbf{k}) \delta_g(\mathbf{k}') \rangle = \delta_{\mathbf{k},-\mathbf{k}'} [P_g(k) + P_N] \tag{248}$$

$$\begin{aligned} \delta_g^{obs}(\mathbf{x}_i) &= \tilde{W}(\mathbf{x}_i) \delta_g(\mathbf{x}_i) \\ \delta_g^{obs}(\mathbf{k}_i) &= \sum_{\mathbf{k}_j} \tilde{W}(\mathbf{k}_j) \delta_g(\mathbf{k}_i - \mathbf{k}_j) \end{aligned} \tag{249}$$

$$\tilde{W}(\mathbf{x}_i) = \begin{cases} 1, & \mathbf{x}_i \, in \, surveyed \, volume \\ 0, & otherwise. \end{cases} \tag{250}$$

A Basic Math

A.1 Special Functions

Legendre:

$$\mathcal{P}_l(\hat{n}\cdot\hat{n}') = \frac{4\pi}{2l+1}\sum_{m=-l}^{l} Y_{lm}(\hat{n})Y^*_{lm}(\hat{n}') \tag{251}$$

$$\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu)\mathcal{P}_{l'}(\mu) = \frac{1}{2l+1}\delta_{ll'} \tag{252}$$

Spherical Harmonics:

$$\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]Y_{lm}(\theta,\phi) = -l(l+1)Y_{lm}(\theta,\phi) \tag{253}$$

$$\int d\Omega Y_{lm}(\hat{n})Y^*_{l'm'}(\hat{n}) = \delta_{ll'}\delta_{mm'} \tag{254}$$

Spherical Bessel Harmonics:

$$\frac{d^2j_l}{dx^2} + \frac{2}{x}\frac{dj_l}{dx} + \left[1 - \frac{l(l+1)}{x^2}\right]j_l = 0 \tag{255}$$

$$\int_0^\infty drr^2 j_l(kr)j_r(k'r) = \frac{\pi}{2k^2}\delta_D(k-k') \tag{256}$$

$$e^{i\mathbf{k}\cdot\mathbf{x}} = \sum_{l=0}^{\infty} i^l(2l+1)j_l(kx)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{x}) \tag{257}$$

$$\frac{1}{2}\int_{-1}^{1} d\mu\mathcal{P}_l(\mu)e^{iz\mu} = \frac{j_l(z)}{(-i)^l} \tag{258}$$

Bessel Function:

$$J_n(x) = \frac{i^{-n}}{\pi}\int_0^\pi d\theta e^{ix\cos\theta}\cos(n\theta) \tag{259}$$

Cosine and sine integrals:

$$\begin{aligned}\mathrm{Ci}(x) &= -\int_x^\infty \frac{\cos z}{z}dz,\\ \mathrm{Si}(x) &= \int_0^x \frac{\sin z}{z}dz.\end{aligned} \tag{260}$$

A.2 Transformations

Fourier:

$$\begin{cases}\Theta(\mathbf{x}) = \int \dfrac{d^3\mathbf{k}}{(2\pi)^3}\Theta(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}}\\ \Theta(\mathbf{k}) = \int d^3\mathbf{x}\Theta(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}}\end{cases} \tag{261}$$

Legendre:

$$\begin{cases} \Theta(k,\hat{k},\hat{p}) = \sum_l (-i)^l (2l+1)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})\Theta_l(k) \\ \Theta_l(k) = \frac{1}{(-i)^l}\int_{-1}^{1}\frac{d\mu}{2}\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})\Theta(k,\hat{k},\hat{p}) \end{cases} \tag{262}$$

Harmonic in real space:

$$\begin{cases} \Theta(\mathbf{x},\hat{p}) = \sum_{l=1}^{\infty}\sum_{m=-l}^{l} a_{lm}(\mathbf{x})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{x}) = \int d\Omega\Theta(\mathbf{x},\hat{p})Y^*_{lm}(\hat{p}) \end{cases} \tag{263}$$

Harmonic in k space:

$$\begin{cases} \Theta_l(\mathbf{k}) = \frac{1}{(-i)^l(2l+1)\mathcal{P}_l(\mu = \hat{k}\cdot\hat{p})}\sum_{m=-l}^{l} a_{lm}(\mathbf{k})Y_{lm}(\hat{p}) \\ a_{lm}(\mathbf{k}) = \int d^3\mathbf{x}a_{lm}(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}} \end{cases} \tag{264}$$

References

[1] Scott Dodelson. *Gravitational lensing.* Cambridge University Press, 2017.

[2] Scott Dodelson and Fabian Schmidt. *Modern Cosmology.* 2020.